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ProbSAP: A comprehensive and high-performance system for student academic performance prediction

Xinning Wang[a], Yuben Zhao[a], Chong Li[a,*], Peng Ren[b]

[a] Ocean University of China, Qingdao, 266000, China
[b] China University Of Petroleum, Qingdao, 257099, China

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ABSTRACT

The student academic performance prediction is becoming an indispensable service in the computer supported intelligent education system. But conventional machine learning-based methods can only exploit the sparse discriminative features of student behaviors in imbalanced academic datasets to predict student academic performance (SAP). Furthermore, there is a lack of imbalanced data processing mechanisms that can efficiently capture student characteristics and achievement. Therefore, we propose a comprehensive and high-performance prediction framework to probe SAP characteristics (ProbSAP) on massive educational data, which can resolve imbalanced data issue and improve academic prediction performance for making course final mark prediction. It consists of three main components: collaborative data processing module for enhancing the data quality, scalable metadata clustering module for alleviating the imbalance of academic features, and XGBoost-enhanced SAP prediction module for academic performance forecasting. The collaborative data processing module integrates multi-dimensional academic data, which sustains a good supply for clustering and modeling in the ProbSAP framework. The comparative evaluation results demonstrate that ProbSAP delivers superior accuracy and efficiency improvement for the course final mark prediction of college students over other state-of-the-art methods such as CNN, SVR, RFR, XGBoost, Catboost-SHAP, and AS-SAN. On average, ProbSAP reduces the mean absolute error (MAE) by 84.76%, 72.11%, and 66.49% compared with XGBoost, Catboost-SHAP, and AS-SAN, respectively. It also leads to a better out-sample fit that minimizes prediction errors between 1% and 9% with over 98% of actual samples.

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1. Introduction

Educational data mining (EDM) has proven itself as a viable paradigm for a wide variety of educational applications to explore massive data from educational settings [1]. It incorporates state-of-the-art techniques to discriminate learners' characteristics and understand their achievements during different stages of the teaching-learning cycle. To overcome students' problems and intervene their behaviors in universities, student performance analysis is implemented to effectively help universities evolve the intervention strategies [2]. Therefore, EDM technology is applied to explore the relationship between college students' acquisition of future achievements and accumulated academic data. As an effective metric to evaluate student achievement, student academic performance (SAP) not only serves as a valid means for instructors and academic managers to inspect their strategies in the subsequent semesters, but also allows interviewers and other academic leaders to achieve an overall evaluation of the students in undergraduate or graduate courses [3]. The students' grade point average (GPA) or academic degree represents the student academic performance. SAP prediction that takes advantage of the EDM technology commonly contains academic data mining, future performance prediction and visual learning data analysis in the real educational environment. To discover more hidden knowledge and patterns of the students, diverse educational data accumulated in student information systems, including student genders, ages, assignments and semester examinations [4], have been used to evolve the academic status of each student in different courses. Furthermore, such massive educational data are analyzed to facilitate SAP prediction through machine learning and statistical methods. The majority of SAP prediction techniques including but not limited to classification, clustering and association analysis techniques [5] have been applied to improve student's performance

* Corresponding author.
E-mail addresses: wangxinning@ouc.edu.cn (X. Wang), zhaoyuben@stu.ouc.edu.cn (Y. Zhao), lichong7332@ouc.edu.cn (C. Li), pengren@upc.edu.cn (P. Ren).

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and teaching quality through taking early actions. To further enhance the performance of SAP prediction for heterogeneous data, some researchers have adopted machine-learning-based and deep-learning-based methods for representing students' academic characteristics [6].

Ideally, an accurate failure rate is required to help instructors or professors supply students with personalized recommendations. Machine-learning-based data mining models analyze students' academic performance by calculating transition probabilities of Markov decision process [7], analyzing student's learning behaviors with matrix factorization [8], and extracting multiple features through AdaBoost assembly algorithms, SVM, and neural network [9]. However, such methods generally perform with low prediction accuracy and efficiency when dealing with the class imbalance issue with an inductive bias [10]. Especially under certain conditions such as uneven distribution of low-dimensional educational data, a too strict or loose threshold representing different example boundaries can sift relevant samples or introduce irrelevant samples [11]. Hence, to select an effective threshold is significant for the two-class learning approaches. Furthermore, these existing machine-learning-based strategies cannot precisely dig out the student behavior characteristics while investigating on the sparse or imbalance academic and behavioral data. A few systematic procedures are feasible for SAP prediction even if deep-learning-based data mining technologies have become successful in many significant application fields such as emotion recognition, performance classification, learning analytics, dropout prediction, question difficulty prediction and knowledge tracing [12,13]. Deep-learning-based prediction models such as long short-term memory (LSTM) [14] and convolutional neural network (CNN) [15] are acknowledged to be suitable for large data classification. However, they have no algorithmic strategy to handle the imbalanced curriculum data. Generally, with respect to prediction performance on student behavior data, a balanced class distribution with size ratio of 1:1 can perform comparatively well but does not necessarily achieve the optimal prediction result because the relationship of class distribution and prediction performance is not stated explicitly. Moreover, such aforementioned deep-learning-based prediction methods still rely on the complex neural network with large datasets, leading to the increase of computing time and space complexity. Hence, an efficient technique is required for SAP prediction to deal with the class imbalance and time-consuming issues.

In addition, university-level datasets in different educational settings are generally multi-dimensional. Such datasets usually indicate a multi-dimensional representation and create a computationally expensive deep learning model. However, these deep learning algorithms demand more memory storage and higher computing power to understand and represent characteristics of student behaviors. Most existing deep learning models encounter efficiency, accuracy and scalability issues and small imbalanced datasets lead to the poor prediction performance. Moreover, it is a nontrivial issue to interpretably track each feature importance and the relationship between different features for machine-learning-based SAP prediction. Compared to deep-learning-based prediction models which lack interpretability of how multi-dimensional data are transformed into desired outputs, tree ensemble models like XGBoost [16,17] rarely depends on specific datasets and more efficiently avoid the drop-in performance by decision rules [18]. Nevertheless, they cannot tackle the imbalanced data issue across data resampling or algorithm optimization. Especially for SAP prediction, XGBoost performs better on the balanced data but obviously worse on other imbalanced datasets in Fig. 2(a), and outperforms deep learning methods on different datasets. An effective SAP prediction system also needs to provide a mechanism for diverse EDM applications to efficiently process imbalanced data without costly calculation. In this research, we propose a comprehensive and high-performance prediction framework to probe SAP characteristics (ProbSAP). Note that SAP prediction indicates the course final mark prediction of college students. ProbSAP considers the local correlation and sample similarity between neighboring points. To cope with the challenges from the aforementioned researches, we design several techniques in ProbSAP including collaborative data processing, scalable metadata clustering, and XGBoost-enhanced SAP prediction. We optimize the loss function by second-order Taylor and utilize a number of datasets to evaluate the performance of ProbSAP and validate the effectiveness of our design through the analysis of diverse results. In summary, our research makes the following contributions.

(1) We evaluate the effectiveness of traditional SAP prediction methods for imbalanced educational data and present the problem in existing SAP prediction technologies. To satisfy the need of multiple feature selection and accurate representation, we orchestrate multi-source data management such as the extraction of previous academic performance, the compression of daily activities, and the combination of teachers' teaching information.

(2) We propose a novel student academic prediction technique, namely ProbSAP, which includes collaborative data processing for efficiently integrating multi-dimensional academic data, scalable metadata clustering for imbalanced data management, and XGBoost-enhanced SAP prediction to support precise and fast final mark prediction of college students.

(3) We evaluate the effectiveness and efficiency of ProbSAP for predicting final marks with a broad set of real-world features. Our experimental results demonstrate that ProbSAP can improve the prediction performance by a lower MAE of 1.4219 and outperform two EDM methods, Catboost-SHAP [19] and AS-SAN [20], by 27.89% and 33.51%, respectively. In addition, ProbSAP achieves high efficiency and effectiveness in terms of multi-source imbalanced educational data for the prediction of final marks.

The rest of the paper is structured as follows. Section 2 briefly reviews traditional machine learning methods, deep learning methods, and ensemble methods. Section 3 presents a comprehensive study on the imbalanced data issue. Section 4 introduces the design and implementation of our ProbSAP system in detail. Experimental results are discussed in Section 5. Finally, Section 6 concludes this article and draws some suggestions for future research.

2. Related work

There has been a number of proposals for EDM techniques to improve the SAP prediction performance. We mainly review some latest researches within the context of SAP prediction methods, including traditional machine-learning-based data mining methods, deep-learning-based data mining methods and ensemble-learning-based data mining methods in this section.

2.1. Traditional machine-learning-based data mining methods

Classical EDM encompasses the off-the-shelf machine learning methods such as logistic regression, naive Bayes, support vector machine (SVM), decision tree (DT), and so on, to train prediction models through previous academic behaviors [6,21]. López-Zambrano et al. [22] analyzed the most commonly-used EDM methods such as J48, RF, SVM, and Naive Bayes for classification tasks, and examined the methods of logistic regression and linear regression for regression tasks. Burgos et al. [23] proposed the logistic regression models for classification to predict whether students would drop out of a course. But they just designed the prediction model according to a small quantity of E-learning students and the effectiveness of logistic regression method to reduce dropout rates was not demonstrated. Hussain et al. [6] also compared several different traditional machine learning algorithms, using

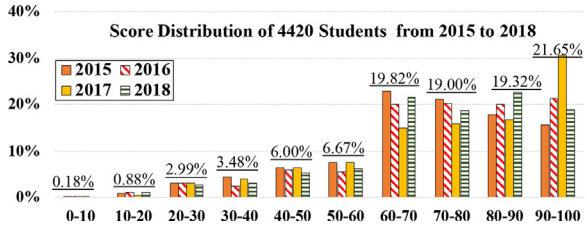


Fig. 1. Score distribution of 4420 students for Probability Statistics course from 2015 to 2018.

data from student activities in class as input variables to predict their grades. In this study, neural network and SVM obtained the higher accuracy than LR, Bayes, and DT classifiers. Alharbi [24] proposed a rule-list algorithm for predicting multi-class performance, and illustrated the importance of interpretable models while predicting student academic performance. In addition, Francis and Babu [25] proposed a prediction model based on the combination of classification and clustering algorithms, and the experimental results demonstrated that the hybrid algorithm outperformed the single machine learning method. Lara et al. [26] analyzed the learning data in electronic environments according to knowledge discovery in database (KDD) and established reference models for students that dropped out of school. However, those traditional academic performance prediction models just focus on the natural distribution of student achievements and ignore the characteristic of sparseness in feature space from imbalanced data [27]. As an example, consider the student performance prediction in curriculum scores as possibly failed. A normal dataset might include 98% passed instances and 2% failed cases in Fig. 1, which leads to a predictive accuracy of 98% for the majority of subjects. Therefore, we try to mitigate the impact of data imbalance on large-scale data and achieve accurate performance and interpretable analysis.

2.2. Deep-learning-based data mining methods

Deep-learning-based data mining algorithms have been widely used as an effective EDM technology by some researchers to predict the student achievement. Liu et al. [14] proposed EKT, a framework using bidirectional LSTM with Markov property and attention mechanism to track student's knowledge situations based on exercising records of 84,909 students, so as to predict student achievements in time series. Waheed et al. [28] employed deep ANN to predict student's risk of failure from the click-stream data, and demonstrated that students who preferred to take certain lectures performed better than others. Su et al. [29] proposed a student performance prediction framework according to regression model and back-propagation (BP) neural network, to solve the cold start problem. Wang et al. [20] proposed AS-SAN model via adaptive sparse self-attention networks, which achieved fine-grained learning performance prediction. In the above studies, LSTM-based SAP forecasting method still depends on feature extraction and sequential computation to enhance prediction accuracy [30], which incurs large runtime overhead for large-scale educational data. Data-driven LSTM models recursively deal with sequential data points and memorize the previous information in order to contribute to the consequent prediction. Nevertheless, the data-driven structures are all based on black-box deep learning models, and they typically construct a complex prediction network with low-dimensional representation. CNN-based prediction techniques not only require large amount of related data to train a forecasting model, but also suffer from architecture-level disadvantages including heavy hardware and calculation costs, excessive parameters, and unessential data transfer [31]. Although the existing prediction models in [32] investigate and examine data-level and algorithm-level solutions for student performance prediction with new loss functions or imbalanced ratios, their methods not only retain the original complicated network structures, but cannot achieve high prediction accuracy with different imbalanced data.

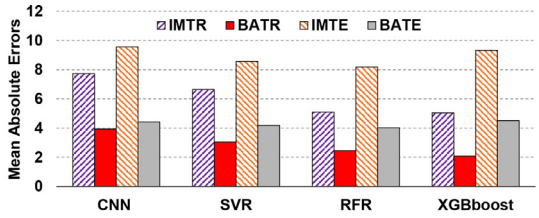
2.3. Ensemble-learning-based data mining methods

Ensemble-learning-based methods have been adopted by multiple proposals to forecast student achievements such as random forest (FR), gradient boost decision tree (GBDT), and extreme gradient boosting (XGBoost). Batool et al. [33] adopted random forest algorithm to explore the impact on academic performance of demographic features, including individual student behavior and family members status.

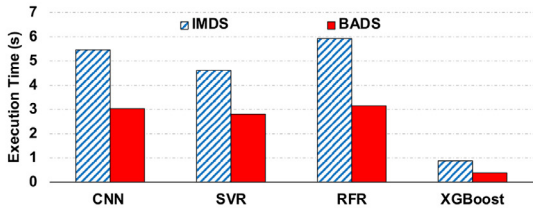
Xu et al. [34] proposed a two-tier prediction structure containing multiple basic models and cascading integration methods, and used matrix decomposition to find correlations between different courses. Their simulations based on undergraduate data demonstrated that the structure could extract the states of student progress and expressed the impact of course information for the whole prediction program. However, these strategies have some shortcomings for dealing with imbalanced academic data. For example, resampling methods caused critical information loss or overfitting, and SMOTE method generated duplication between different categories [35]. Zhai et al. [19] proposed K-prototype-based student portrait construction and Catboost-SHAP-based academic prediction, which provided high-performance ensemble learning methods with interpretable analysis. The performance of ensemble EDM methods demonstrates its effectiveness of handling textual sentiment data, which motivates us to construct a novel ensemble predictive model for multi-source data. Furthermore, we propose the fusion framework based on clustering and extreme gradient boosting to alleviate the imbalance while keeping the original data unaltered. In addition, the above studies show how student performance is linked to certain factors in a single aspect. Nevertheless, our work focuses on both chronology and comprehensiveness, that is, taking into account student performance in both the historical years and the current semester, as well as comprehensive characteristics that may influence academic performance.

3. Motivation

In this section, we evaluate student grade prediction performance to understand behavior features from original learning data shown in Fig. 1 in Probability Statistics course for 4420 students from 2015 to 2018. Obviously, the student score dataset include approximately 20% failed cases ranging from 0 to 59 points and 80% passed cases ranging from 60 to 100 points. The score distribution is imbalanced for practical EDM applications. Note that the item *score* indicates the course final marks in the study. The least students (smallest ratio of 0.18%) achieved a score of less than 10, but more than 20% of students (largest ratio of 21.65%) achieved a score between 90 and 100 in Probability Statistics course. We utilize four prediction algorithms including traditional machine-learning-based method, deep-learning-based method, and ensemble prediction methods. For each method that we use for data mining and performance evaluation, the mean absolute error (MAE) and execution time are compared and discussed. Neural network learning method such as CNN [36] extracts individual characteristics through multiple computing layers. Mathematical regression learning technique such as SVR [37] uses kernel functions to obtain multi-dimensional features that best fits the separating hyperplane within multiple separable datasets. Random forest regression model (RFR) [38] uses bootstrap aggregating (i.e., bagging and averaging) to skip some observations for successful model training.



(a) Mean absolute error of CNN, SVR, RFR, and XGBoost with imbalanced training set (IMTR), balanced training set (BATR), imbalanced testing set (IMTE) and balanced testing set (BATE).



(b) Execution time of CNN, SVR, RFR, and XGBoost with imbalanced dataset (IMDS) and balanced dataset (BADS).

Fig. 2. Mean absolute error and execution time of CNN, SVR, RFR, and XGBoost with balanced and imbalanced sets.

Boosting ensemble learning technique such as XGBoost [16] provides scalable tree boosting strategies for sparse data.

In view of the severe data imbalance as shown in Fig. 1, we have further examined the impact of imbalanced score data in Fig. 2(a). We build class-imbalanced sets and class-balanced sets with the samples in the ranges of 10–20, 80–90, 70–80 and 80–90, and categorize them into imbalanced training set (IMTR), balanced training set (BATR), imbalanced testing set (IMTE), and balanced testing set (BATE) according to the ratio of 8:2. Fig. 2(a) shows the MAE comparison of CNN, SVR, RFR and XGBoost. For all the data, XGBoost with BATR achieves the lowest MAE with 2.08 points than other prediction methods due to its ensemble learning strategy that can reduce the bias and variance compared to the single learner, but it incurs overfitting with a higher error for testing sets of IMTE and BATE. Note that deep learning method CNN cannot provide accurate prediction performance by larger MAEs than SVM, RFR, and XGBoost because complex neural layers hardly extract student features from imbalanced data. In contrast, RFR outperforms normal deep learning and machine learning methods such as CNN and SVR. This is because RFR integrates multiple learners and averagely computes predicted values to alleviate the gap between inputs and outputs. As we can see from this figure, XGBoost with IMTR and BATR obtains a smaller MAE compared to RFR. However, baseline XGBoost with IMTE and BATE suffers from overfitting for similar testing procedures.

Fig. 2(b) shows execution time of CNN, SVR, RFR, and XGBoost with imbalanced dataset (IMDS) and balanced dataset (BADS). Due to parallel and distributed computing, XGBoost can efficiently predict student course marks just in 0.87s and 0.39s for IMDS and BADS respectively. To ensure the relative independence of each learner, RFR focuses for a long time on building multiple trees and calculating weight values. Hence RFR scheme delivers the highest time consumption, for example, 5.93s for IMDS and 3.17s for BADS respectively. Since balanced data affects computation complexity of score prediction, as shown in Fig. 2(b), the time consumption drops to 40% $\sim$ 56% among four prediction methods with balanced datasets (BADS). Although XGBoost delivers a fast execution speed and a small prediction error for balanced data, it achieves the largest MAE for imbalanced testing data compared to CNN, SVR, and RFR in Fig. 2(a). To alleviate the influence of class imbalance for academic data mining and student final mark prediction, we propose a new prediction framework ProbSAP, which systematically learns student academic characteristics.

4. ProbSAP system

As described in the previous section, imbalanced data lead to large MAEs for student final mark prediction, especially severe overfitting in training datasets. With more characteristics not being understood in CNN, SVR, RFR, and XGBoost, the space size of learning representation is significantly reduced for imbalanced categories. To address this problem, we present the comprehensive and high-performance (ProbSAP) framework, including collaborative data processing, scalable metadata clustering, and XGBoost-enhanced SAP prediction. Based on the observation, ProbSAP aims to realize multi-dimensional EDM framework, tackle data imbalance from data and algorithm levels, and improve the generalization performance of prediction methods. In doing so, ProbSAP can alleviate MAEs and computation overhead such that more accurate scores can be predicted for imbalanced data.

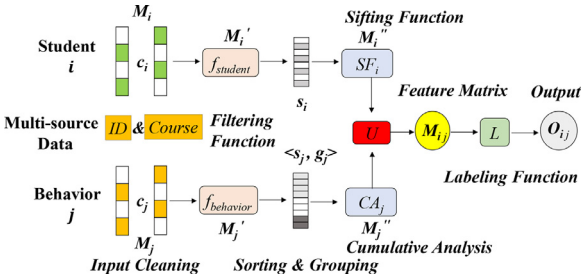


Fig. 3. Collaborative data processing framework with six parts: input cleaning, filtering function, behavioral features sorting & grouping, student features sifting function, multiple data cumulative analysis, and feature matrix labeling.

4.1. Collaborative data processing

1) Input cleaning: The multi-dimensional distribution from Dean Office SQL database is used to construct an imbalanced classification and regression dataset, which contains approximately 108,034 college students, 206,314 examination score records, and more than 48 million campus consumption records in Table A.1 in the appendix which describes data distributions with detailed information on multiple features and sample amount from GitHub[1] and Figshare.[2] To enhance the data quality in collaborative data processing, we separate original data into three categories including student information, behavior representation, and others. For simplicity, each data sample is divided into two multi-dimensional vectors such as *Student i* and *Behavior j* in Fig. 3 and other relatively important data in Table A.1 are selected as supplementary attributes and incorporated to *Behavior* vector. In general, student information consists of elemental features created by students such as student ID, student name, age, hometown, gender, and so on. Behavior representation includes plenty of course attributes from administration system, exam histories from teaching system, scholarship records from administration system, and campus activities from logistics system. Due to data redundancy and deficiency in reality, an efficient data cleaning mechanism is required to handle irrational educational data. Specifically, we remove redundant data

[1] https://github.com/mulanbenben/Student_academic_data.
[2] https://figshare.com/articles/dataset/Student_academic_data/21685748.

Table 1
Basic features and calculated behavior features in output matrix O_{ij} for the final mark prediction of Probability Statistics.

Feature	Description	Range
SID	Student ID	010012003001–212552018399
ACS	Advanced Calculus score	0–100
ILAS	Introduction to Linear Algebra score	0–100
AMS I	Advanced Mathematics I score	0–100
AMS II	Score of Advanced Mathematics II score	0–100
CFITS	Complex Function and Integral Transformation score	0–100
AS	Average score	45.2–95.5
CGPA	Cumulative grade point average	0–5.0
GPA_1	Grade point average in the 1st academic year	0–5.0
GPA_2	Grade point average in the 2nd academic year	0–5.0
GPA_3	Grade point average in the 3rd academic year	0–5.0
TFP	Numbers of times student failed an exam	0–5
ASPS	Average score of teachers in Probability Statistics	0–100
BF	Breakfast frequency in current semester	0–120
BT	Breakfast time in current semester	0–120
ASCC	The amount spent in campus card in current semester	1860.5–4428.7
PSS	Probability Statistics score	0–100

records and incomplete samples and modify the original data format for further data management.

Input cleaning stands for the input of student and behavior, including ID-based identifying, course-based replacing, and daily-behavior-based scrubbing. It indicates a student with relative academic behaviors or a course with interacted attributes i.e., one row or one column in student-behavior interaction matrix. The definitions are shown as follows:

$$\begin{aligned} c_i &= \mathbb{ID}(i)\ and\ course(i)\ and\ behavior(i), \\ c_j &= \mathbb{ID}(j)\ and\ course(j)\ and\ behavior(j), \end{aligned} \tag{1}$$

where c_i, c_j represent the input cleaning results and i, j are the ith row and jth column of student-behavior matrix. Additionally, $ID()$ refers to identifying irrelevant student attributes according to the unique identity. $course()$ means the data replacing of invalid courses based on approximate mean of curriculum courses and $behavior()$ stands for the modifying or deleting of coarse behaviors. Then the clean data matrices without fine-tuning c_i and c_j are obtained for the corresponding data processing.

2) Filtering function: To examine the extraction of the relevant features for mark prediction, we filter off gender, age, social security ID, telephone number, parent information, etc., through the filtering function $f_{student}$ as follows:

$$M_i' = f_{student}\left\{c_{i\ (gender,\ age,\ social,\ tel,\ parents,\ldots)}\right\}. \tag{2}$$

Equation (2) converts the input cleaning c_i to student attribute filtering M_i'. However, there are more inconsistent data instances in multi-source behavior matrix c_j because some students failed the compulsory courses and restudied the same course to finish the study plan. Therefore, we screen out the duplicate columns by filtering function $f_{behavior}$ for the same courses that students repeatedly selected as follows:

$$M_j' = f_{behavior}\left\{c_{j\ (course_selection)}\ or\ c_{j\ (daily_life_record)}\right\}. \tag{3}$$

To effectively track the student achievement about how many credits they have obtained, we just consider the final results instead of intermediate results according to course selection situation ($c_{j\ (course_selection)}$). In addition, the daily life records such as consumption frequency and activity venue also naturally reflect student's study habits and provide more accurate prediction features that can influence discriminating performance and efficiency by specific daily life filtering ($c_{j\ (daily_life_record)}$) of Eq. (3).

3) Sorting & grouping: The elements of student matrix M_i' by Eq. (2) contain enrollment time and grade level with a similar format. For example, in student ID *15013***2006**023, *15013* and **2006** represent the department number and enrollment time as well as indicate the grade level. To understand the student distribution more easily, we sort student matrix by a portion of student ID such as department number and enrollment year. The students in the same grade generally choose the similar courses and participate in similar activities. Hence, we first organize the student matrix by enrollment time in ascending order and store it for optimum efficiency. With respect to selected courses, each course ID also contains the department information that the course belongs to. If we sort the courses according to the student final marks, the implications of mark variation may be adjacently shown in the sorted elements. Moreover, to efficiently find the relevant courses and extract their features, we utilize grouping strategy in behavior matrix M_j'. Take "Probability Statistics" course as an example, after sorting and grouping we can extract some similar attributes based on common characteristics. When we analyze other mathematical courses, e.g., "Advanced Calculus", "Introduction to Linear Algebra", "Advanced Mathematics", and "Complex Variable Functions and Integral Transformations", the common mathematical disciplines from previous extraction are helpful to reduce loop-up time. Furthermore, the related features in Table 1 can serve as the feasible inputs for student achievement learning.

4) Sifting function: To detect useful information and reduce standard deviations, we apply a conventional data sifting method to discard some outliers while retaining the majority of good data. For different level students such as freshman, sophomore, junior, and senior, the knowledge states of a certain student can conspicuously change according to effective knowledge acquisition. For example, the behavior records of senior students that earn more semester hours have higher dimensional academic features than other undergraduate students. Due to the inconsistency of feature space, it is difficult to construct a unified prediction model that is applicable to all kinds of student behaviors. Therefore, we sieve multi-source features to eliminate some noisy information by statistical analysis as below:

$$M_i'' = SF_i\left\{P(\chi^2_{min} \sim d)\ and\ cov(M_i')^{-1} = \frac{\partial'' \chi^2}{\partial \alpha_s \partial \alpha_k}\right\}, \tag{4}$$

where $\chi^2_{\min}$, d, $cov(M_i')^{-1}$, α represent the χ likelihood method, degree of freedom, covariance matrix, and student attributes respectively. When the value $\chi^2_{\min}$ approaches the degrees of freedom denoted as the number of data points minus the number of fitted variables, a reasonably high probability is used to evaluate student development judgement. Furthermore, the covariance

matrix with partial derivatives allows us to compute the correlations between parameter α_s and α_k. Based on the two criteria, sifting method not only provides reasonable features, but statistically evaluates correlations among multiple basic features.

5) Cumulative analysis: To quantify the impacts of historical information, we conduct cumulative analysis such as grade point average (GPA) and cumulative activity expense based on behavior data $M_j^{'}$. For example, the valuable GPA records are nontrivial to represent the learning characteristics from a quantitative perspective. Hence, we build a GPA vector including the cumulative calculations of the first three academic years and the final cumulative GPA (CGPA) by

$$\mathbf{GPA}_i = \left[GPA_i^1, GPA_i^2, GPA_i^3, CGPA_i \right]^T, \tag{5}$$

where $\mathbf{GPA}_i$ is the ith GPA vector, in other words, it stands for a collection of GPA parameters of the ith student. In this GPA vector, GPA_i^1, GPA_i^2, and GPA_i^3 mean the GPAs of the ith sample in first three academic years, and $CGPA_i$ means the CGPA of the ith sample. GPA_i^k and $CGPA_i$ can be calculated by Eqs. (6) and (7)

$$GPA_i^k = \frac{\sum_{j=1}^{Num_i^k} Credit_j \times \left(Score_j/10 - 5 \right)}{\sum_{j=1}^{Num_i^k} Credit_j}, \tag{6}$$

$$CGPA_i = \frac{\sum_{j=1}^{Num_i} Credit_j \times \left(Score_j/10 - 5 \right)}{\sum_{j=1}^{Num_i} Credit_j}, \tag{7}$$

where Num_i^k is the number of courses for the ith student in the kth academic year. Similarly, Num_i is the number of accumulated courses for the ith student since enrollment semester. $Score_j$ and $Credit_j$ are the score and corresponding credit of the jth course. In addition, we calculate student's past average scores, accumulated scholarship, and the number of times the student failed exams to extract multi-dimensional grade-related features.

Specifically, the daily individual behaviors are of great relevance to student academic achievement. But due to its low information density and the large amount of original data, there is a lack of data processing method that can efficiently exploit these features. Therefore, we extract some useful daily personal features such as breakfast frequency, breakfast time, and the money they spent in the first 120 days of each semester, and calculate the average score of courses taught by the same teachers to enrich feature matrix $M_j^{'}$ to $M_j^{''}$.

6) Labeling function: After above collaborative data processing, feature matrix M_{ij} is obtained by unifying the student matrix $M_i^{''}$ and the behavior matrix $M_j^{''}$ through the same student ID. The students in different departments may choose the same course (e.g., Probability Statistics course is also chosen by students who study mathematics, finance, chemistry, etc.), and hence the departments that students choose can serve as an indicator for final mark prediction. However, the grades of students who plan to change their major may cause category classification confusion. Hence, we label the samples according to the first major attributes, course information and the scores in the first academic year. The labeling function L is defined as follows:

$$O_{ij} = L\left\{ M_{ij\ (student_ID)}\ and\ M_{ij\ (course_ID)}\ and\ M_{ij\ (score_first_year)} \right\}. \tag{8}$$

Equation (8) attempts to fine-tune feature matrix M_{ij} to output matrix O_{ij} by labeling multiple samples. For example, the course IDs of Probability Statistics and Advanced Mathematics are **1312** and **1309**. Note that **13** represents mathematics. To discriminate the students who selected the two courses, their student IDs are prefixed with the major information, for instance, *12008-mathematics*, *15204-finance*, and *09112-chemistry*. Labeling function

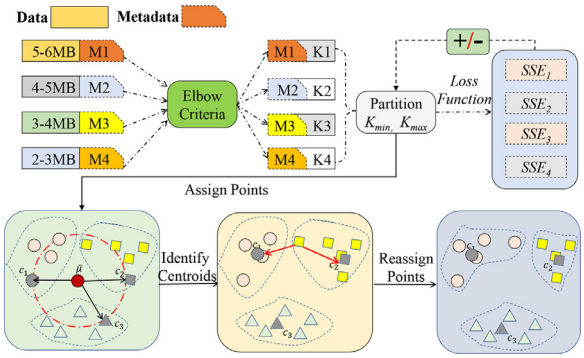


Fig. 4. Scalable metadata clustering framework of ProbSAP for solving data imbalance, which mainly contains scalable metadata structure, partition K-value adjustment, and metadata-based clustering.

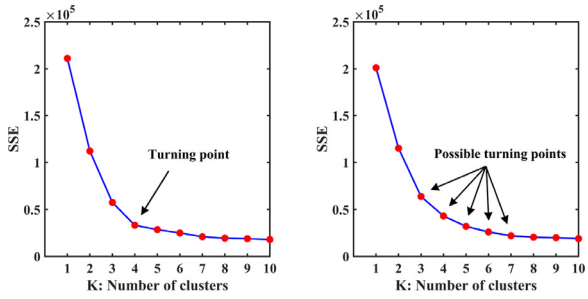
L identifies course categories **13** and then labels the samples into three subcategories according to major information *12008*, *15204*, and *09112*. After that, the scores labeled as different majors are tentatively removed from the dataset used to predict different types of courses. Finally, an output matrix O_{ij} is obtained for the case of the final mark prediction of Probability Statistics through collaborative data processing in Table 1. It retains the basic features from the inputs *Student i* and *Behavior j* and generates the helpful academic characteristics to enhance the data integrity. In detail, take Probability Statistics mark prediction as an example, its output matrix O_{ij} not only includes 7 basic features (e.g., SID, ACS, ILAS, AMSI, AMSII, CFITS, and AS), but also contains 10 calculated features in Table 1 (e.g., CGPA, GPA_1, GPA_2, GPA_3, TFP, ASPS, BF, BT, ASCC, and PSS). Among them, ASPS, BF, BT, and ASCC represent average score of teachers in Probability Statistics, breakfast frequency, breakfast time, and the amount spent in campus card in current semester respectively, which are reconstructed from the daily activities, test scores and teacher's information through collaborative data processing.

4.2. Scalable metadata clustering

In this section, to effectively solve the imbalance problem for multi-dimensional academic data, we design a scalable metadata clustering framework as shown in Fig. 4. This framework includes scalable metadata structure, objects partition adjustment, and metadata-based clustering.

1) Scalable metadata structure: Scalable metadata structure is displayed at the top left of Fig. 4, which is designed to efficiently read and write metadata in the process of clustering. In this structure, the data in matrix O_{ij} is first stored in contiguous memory according to different sample labels and the amount of labeled data. For example, the samples with a size of 3.45MB that are labeled as finance-math (i.e., department-course) can be stored in segment "3-4MB". In order to facilitate fast metadata clustering operations, the metadata segments are transformed into the new format (e.g., M1 and M2, etc.) consisting of cluster centroids and the samples closest to the centroid generated by clustering algorithm (described in Section 4.2 **3**)), represented as $< Cent_i, \{Samp_i^1, Samp_i^2, Samp_i^3, \ldots, Samp_i^n\} >$. $Cent_i$ is the ith centroid position calculated by 2-dimensional features (e.g., AS&BF or CGPA&BF) among the features of matrix O_{ij}, and $Samp_i^j$ is the jth sample with 15-dimensional features in the ith cluster.

In the above data structure definition, since the number of centroids (clusters) K severely affects the results of final mark prediction, we apply Elbow Criteria to discriminate the optimal cluster numbers K by measuring the sum of squares due to error (SSE)



(a) AS&BF-dependent. (b) CGPA&BF-dependent.

Fig. 5. SSE decay curve for determining the optimal cluster number based on Elbow Criteria, which contains two cases: (a) a sharply decreasing curve with the explicit turning point and (b) a fairly smooth curve with the ambiguous turning points.

between centroid $Cent_j$ and each sample $Samp_i^j$, as follows:

$$SSE = \sum_{i=1}^{K} \sum_{Samp_i^j \in Clus_i} \left\| Cent_i - Samp_i^j \right\|^2, \tag{9}$$

where $Cent_i = \frac{1}{|Clus_i|} \sum_{Samp_i \in Clus_i} Samp_i$, i.e., the mean value of all the samples in $Clus_i$. Elbow Criteria is a statistical metric for estimating potential cluster numbers [39]. Theoretically, when the initial optimal value K is less than the final optimal value, the loss function will significantly reduce SSEs for multi-source data samples; when the potential value K is bigger than the optimal value, the loss function will change SSEs slightly.

Fig. 5(a) and (b) show examples of determining the optimal cluster number on different datasets (e.g., AS&BF and CGPA&BF). AS and CGPA represent students' past learning situation, and BF is the behavior pattern in current semester in Table 1. In Fig. 5(a), we try to distinguish the potential value K according to the plateau (i.e., K=4). But for other datasets (e.g., CGPA&BF), the turning point (i.e., elbow point) cannot be unambiguously distinguished when the plotted curve is fairly smooth in Fig. 5(b). Therefore, we design a partition (cluster/centroid) K-value adjustment method to determine the optimal centroid number in Eq. (9) such that our metadata such as M1-M4 of Fig. 4 can directly access the centroids and relevant samples.

2) Partition K-value adjustment: In order to further find appropriate clustering number K for a case like Fig. 5(b), shown from the top right of Fig. 4, we design a refined partition K-value adjustment module. In the initialization phase, we calculate the average distortion AD_k of the kth K-value sets in the initial estimate by computing the average point distortion of K clusters as below:

$$\begin{aligned} AD_k &= \frac{\sum_{j=1}^{K} \left(SSE_j / Num_j \right)}{K} \\ &= \frac{\sum_{j=1}^{K} \left(\sum_{Samp_i^j \in Clus_j} \left\| Cent_j - Samp_i^j \right\|^2 / Num_j \right)}{K}, \end{aligned} \tag{10}$$

where SSE_j represents the total distortion of the jth cluster of Num_j points, i.e., the quadratic sum from its each point $Samp_i^j$ to the jth cluster centroid $Cent_j$, and SSE_j/Num_j is the average point distortion of the jth cluster. To weigh the influence of the K increase on the point distortion, we further calculate the difference D_k of average distortion under the adjacent K values as follows:

$$D_k = |AD_{k+1} - AD_k|, \tag{11}$$

where AD_k and AD_{k+1} represent the average point distortion at the kth and $(k+1)$th K values, and since the difference is always negative, the absolute value is calculated for result comparison. Thus, the distortion incremental vector $\mathbf{D} = [D_1, D_2, \ldots, D_k]$ containing the information around the turning point is obtained by Eq. (11).

For Fig. 5(b), we try to tune the value of K to have the optimal partition. To determine the optimal K from the estimated values (K=4,5,6), we compute the incremental vector $\mathbf{D} = [D_1, D_2, D_3, D_4, D_5, D_6]$ from K=2 to K=8. Through comparing each element in this vector, we observe that compared to the extremely large values of D_1, D_2, and D_3, D_4 are smaller but significantly larger than D_5, D_6. It shows that the K-value increase has a significant impact on clustering before K=4 and the effect becomes smaller after K=6. Therefore, we can determine that the minimum value of the turning point K_{min} is 4, and the maximum value K_{max} is 6. Finally, the range of the possible turning point is further narrowed to K=4, 5, 6 for Fig. 5(b). Those possibly optimal K values obtained by the vector **D** will be stored to the scalable metadata structure such that multiple processes in the application can be parallelized to quickly access all the locations of K clusters and corresponding samples when training models and predicting course final marks. The prediction effects of different K values will be verified and discussed in Section 5.2.

3) Metadata-based clustering:

Shown from the bottom of Fig. 4, a metadata-based clustering module is designed to flexibly find the optimal equilibrium of the input feature matrix by iteratively assigning samples to different categories. The assignment is based on the real feature distribution for alleviating the imbalance of academic data. In the beginning, the input points from different partitions are assigned to K clusters which are initialized by our partition K-value adjustment method. It is proved that the more scattered centroids more effectively preserve the clustering performance. Hence, we initialize the centroids based on equivalent distance to the average center point $\overline{\mu}$ for reducing the computation overhead. Particularly, the center position $\overline{\mu}$ of all the training points is calculated as follows:

$$\overline{\mu} = \frac{1}{Num} \sum_{i=1}^{Num} (Samp_{i1}, Samp_{i2}), \tag{12}$$

where Num is sample size, $Samp_{i1}$ and $Samp_{i2}$ are two-dimensional features (AS and BF, or CGPA and BF). $\overline{\mu}$ is selected as circle center and radius keeps the balanced distribution of inside circle and outside the circle as follows:

$$N_1 = Num(dist(\overline{\mu}, x_i) < r) \text{ and } N_2 = Num(dist(\overline{\mu}, x_i) > r). \tag{13}$$

After identifying all the centroids and the distance between each point x_i to its center $Cent_i$, we try to adjust the points to the correct cluster whose centroid distance is minimum. For the sub-partitions, K-means clustering method recalculates its centroids (i.e., the new centroid for the samples in the sub-partition) as follows:

$$Cent_i = \frac{1}{|Clus_i|} \sum_{Samp_i \in Clus_i} Samp_i. \tag{14}$$

And then we calculate the distance between the points and the new centroids and reassign points to the new category where the centroid with the shortest distance is located. Meanwhile, we repeat the above operations to update centroids and recurrently assign misclassified points until all the centroid positions no longer change.

Fig. 6 shows the centroid variation when the number of calculated clusters increases from 4 to 6. The abscissa and ordinate values are the BF and AS. When validating clustering performance for 20 iterations, the process achieves an equilibrium and stabilization for the centroid distribution. Empirically, the previous scalable metadata strategy gives multi-source imbalance samples an efficient convergence of iterative computation with K=4,5,6 (e.g., the positions of 4, 5, 6 clusters change slightly after 2 iterations in Fig. 6). Similar to Figs. 6, 7 compares the clustering results with different features (CGPA and BF) and K-values (i.e., the number of

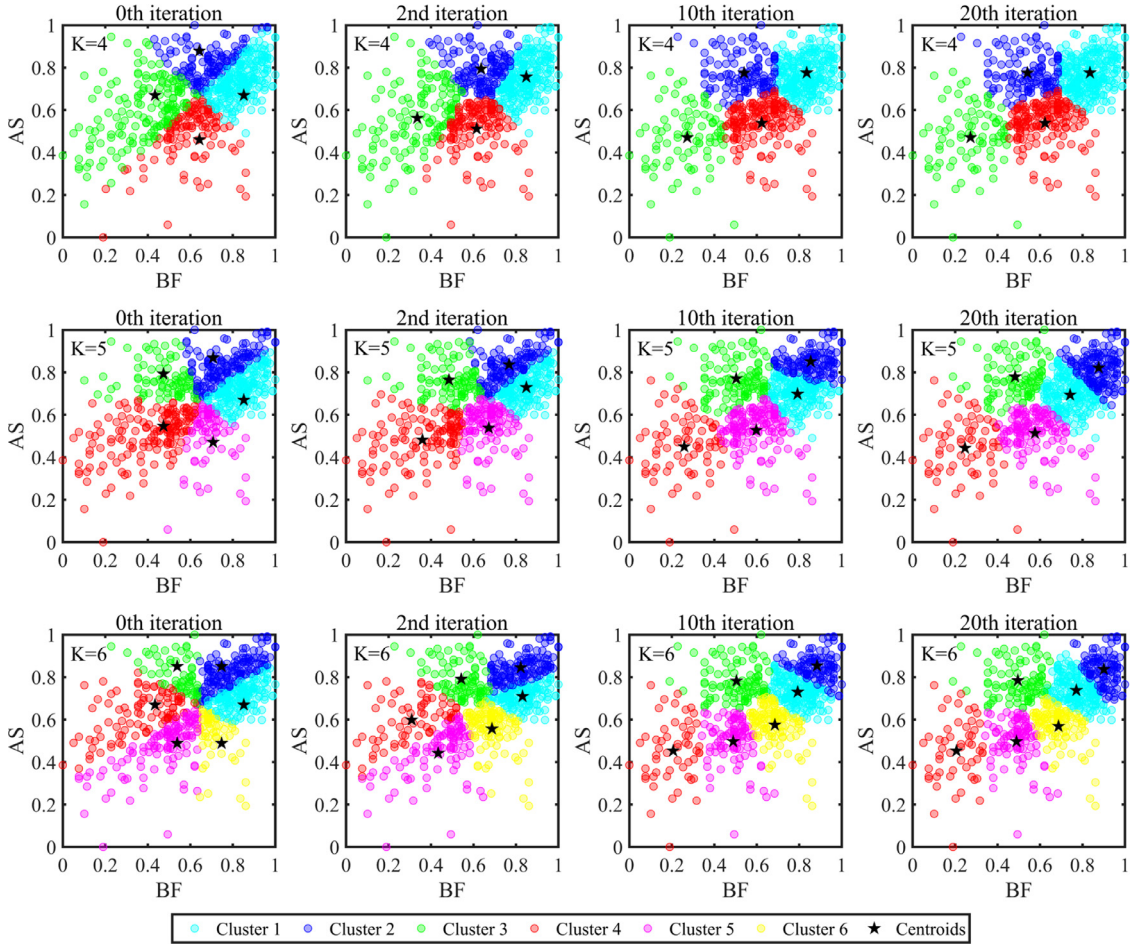


Fig. 6. Centroid variation with AS&BF features when the number of calculated clusters K increases from 4 to 6.

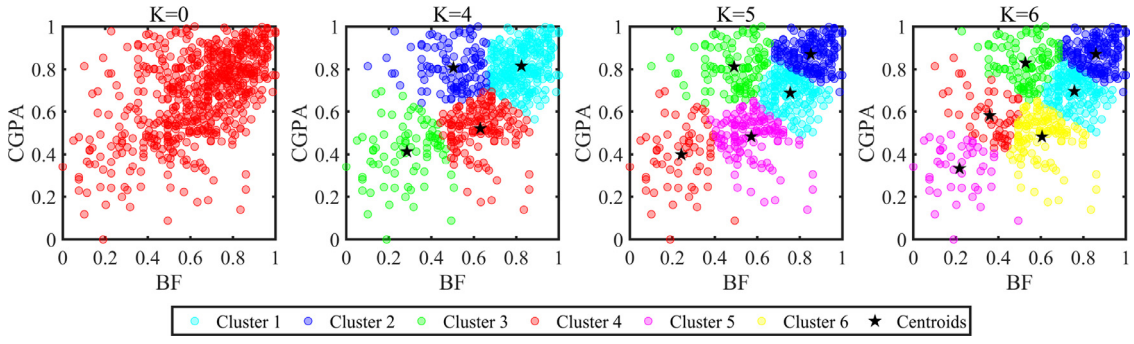


Fig. 7. Comparison of clustering results and data distribution with CGPA&BF features under different K values.

clusters is 4, 5, and 6). We can see that the number of clusters incurs representative partitions. There is a declining tendency of inter-cluster distance for 4 clusters up to 6 clusters. The practical influence on prediction performance will be discussed detailedly in Section 5.2.

4.3. XGBoost-enhanced SAP prediction

This section introduces the XGBoost-enhanced SAP prediction module as shown in Fig. 8, which mainly includes precise residual calculation and XGBoost-enhanced modeling.

1) Precise residual calculation: In order to accurately capture the residual part between actual and fitted values for predictive accuracy enhancement, we define a new loss function for XGBoost algorithm. XGBoost is an improved gradient enhancement library based on gradient lifting by Chen and Guestrin [16], which can continuously improve the fitting effect by generating weak learners based on the residual results. In our course final mark prediction, assuming that we have trained T trees by XGBoost, the ith sample's predictive score $\hat{Scor}_i$ can be represented as below:

$$\begin{aligned}\hat{Scor}_i &= \sum_{t=1}^{T} Fit_t(Samp_i), Fit_t \in \mathcal{F}, \quad \mathcal{F} \\ &= \left\{Fit(Samp) = w_{q(x)}\right\}\left(q : \mathbb{R}^m \to M, w \in \mathbb{R}^M\right),\end{aligned} \tag{15}$$

where $Fit_t(Samp_i)$ represents the predicted values of the tth tree for the ith sample, and $\mathcal{F}$ denotes the space of regression trees, q is the tree structure that maps samples to leaf indices, m is the

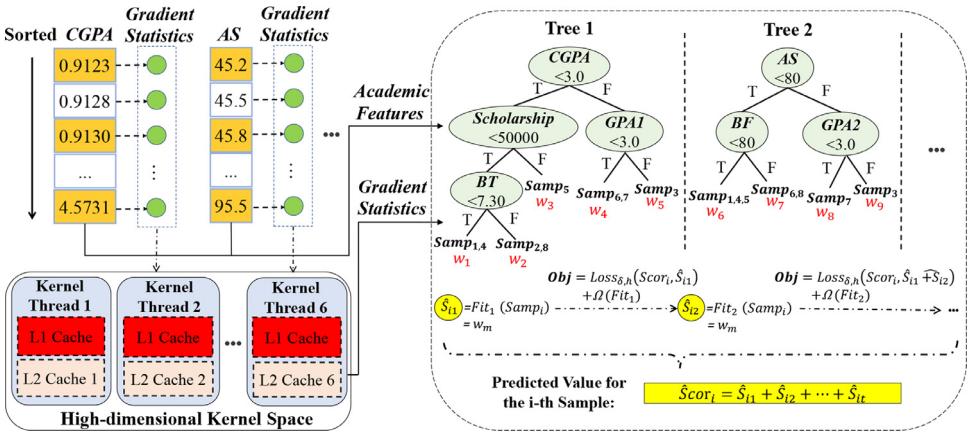


Fig. 8. XGBoost-enhanced SAP prediction framework with two parts: precise residual calculation and XGBoost-enhanced modeling.

number of features, and M is the number of leaf nodes in the current tree. Each Fit_t has its own tree structure q and leaf weight w. The objective function can be described as below:

$$Obj = \sum_{i=1}^{n} Loss\left(Scor_i, \hat{Scor}_i\right) + \sum_{t=1}^{T} \Omega(Fit_t), \tag{16}$$

where $Scor_i$ is sample's actual score, and $Loss\left(Scor_i, \hat{Scor}_i\right)$ is the loss function to measure the residual between predicted and actual values during several iterations of XGBoost. For original XGBoost method, mean square error (MSE) is used in the loss function, but the quadratic term imposes greater penalties for situations with large errors $(Scor_i - \hat{Scor}_i > 1)$ and smaller penalties for situations with smaller errors $(Scor_i - \hat{Scor}_i < 1)$. As a matter of fact, there are two scenarios that should be considered in our academic prediction: (1) Abnormal data: most of student's features are good (poor) but the actual score of the predicted course is low (high). (2) Normal data: all the student's features are consistent with the final scores except for abnormal data.

To reduce the prediction errors for abnormal data, we introduce MAE as an alternative to the loss function. MAE more naturally measures the errors between predicted and actual values. Furthermore, because it has the same penalty for $(Scor_i - \hat{Scor}_i > 1)$ and $(Scor_i - \hat{Scor}_i < 1)$, prediction model may be less sensitive to abnormal samples and its robustness and effectiveness are better than baseline XGBoost. To differentiate MAE, Huber function is introduced:

$$Loss_\delta(Scor_i, \hat{Scor}_i) = \begin{cases} \frac{1}{2}(Scor_i - \hat{Scor}_i)^2, & |Scor_i - \hat{Scor}_i| \le \delta \\ \delta|Scor_i - \hat{Scor}_i| - \frac{1}{2}\delta^2, & \text{otherwise} \end{cases}, \tag{17}$$

where δ is the super parameter which controls linear and secondary errors switching. When the error is in $(-\delta, +\delta)$, Huber loss is equivalent to MSE. But while the error is in $(-\infty, \delta]$ and $[\delta, +\infty)$, Huber loss is equivalent to MAE. To facilitate the derivation, it is smoothed by Eq. (18):

$$Loss_\delta\left(Scor_i, \hat{Scor}_i\right) = \delta^2\left(\sqrt{1 + \left((Scor_i - \hat{Scor}_i)/\delta\right)^2} - 1\right). \tag{18}$$

However, Huber is an absolute error virtually and converges more slowly than MSE. To further speed up the convergence of modeling, we propose S-Huber, which adds the square term on the basis of Huber in Eq. (19):

$$Loss_{\delta,h}\left(Scor_i, \hat{Scor}_i\right) = \delta^2\left(\sqrt{1 + \left((Scor_i - \hat{Scor}_i)/\delta\right)^2} - 1\right) + h(Scor_i - \hat{Scor}_i)^2, \tag{19}$$

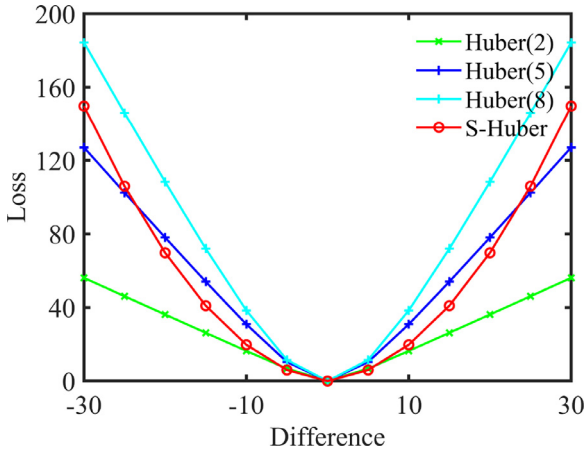


Fig. 9. Comparison of loss values with different input parameters δ=2, 5, 8 for Huber and δ=0.5, h= 0.15 for S-Huber.

where h is the weight coefficient of the square term. Fig. 9 compares the loss values with different parameters. For instance, δ is set to 2, 5, 8 in Huber and δ is set to 0.5 with h=0.15 in S-Huber. It can be observed from Fig. 9 that for normal data with small difference, both the results of Huber and S-Huber are similar to MSE and S-Huber is closer to Huber (δ=2). But when abnormal data with large difference appear, S-Huber must approximate to MAE to lower the contribution of global loss and speed up the convergence (δ=0.5). This shows that Eq. (19) overcomes the sensitivity of MSE to abnormal data and ensures fast convergence of the XGBoost modeling when δ and h are set reasonably.

The left part of Fig. 8 shows the academic features sorting by the column mechanism of XGBoost, which is applied to reduce the computational consumption during modeling. Since repeatedly sorting features during tree split is the most time-consuming, this mechanism performs the data split by scanning the pre-sorted features. However, in high-dimensional kernel level, it needs to access non-contiguous memory in the feature order to obtain gradient statistics. So, to reduce the normal-case access overhead, we allocate a thread for each cluster, and each thread allocates the first-level and second-level caches to store the gradient information, mitigating this problem through cache-aware prefetching.

2) XGBoost-enhanced modeling: To facilitate the deployment of improved loss function in the uniform format of XGBoost, we take the second-order Taylor expansion with respect to the vari-

Algorithm 1 Implementation of SAP prediction framework.

Input: $D = \{Samp_1, Samp_2, \cdots, Samp_{Num}\}$, academic features set

1: // Scalable metadata clustering
2: **for** j = 1 to K **do**
3: $SSE = \sum_{j=1}^{K} \sum_{Samp_i \in Clus_j} \|Samp_i - Cent_j\|^2$
4: **end for**
5: $K \leftarrow$ the turning point of SSE curve
6: $AD_k \leftarrow \frac{\sum_{j=1}^{K} (\mathrm{SSE}_j / \mathrm{NNm}_j)}{K}$, $D_k \leftarrow D_k = |\mathrm{AD}_{k+1} - \mathrm{AD}_k|$
7: $K \leftarrow$ the point of D_k changes slowly
8: $\bar{\mu} \leftarrow \frac{1}{Num} \sum_{i=1}^{Num} (Samp_{i1}, Samp_{i2})$, $r \leftarrow r(N_1 = N_2)$
9: **Cent** $\leftarrow \{Cent_1, Cent_2, \ldots, Cent_K\}$
10: **while Cent** no longer changes **do**
11: **for** i = 1 to Num **do**
12: **for** j = 1 to K **do**
13: $Clus_j \leftarrow Samp_i \; Cent_j \subset Clus_j$
14: **end for**
15: **end for**
16: **end while**
17: // XGBoost-enhanced SAP prediction
18: $Gain \leftarrow 0$
19: $\hat{Scor}_i \leftarrow 0$
20: $G \leftarrow \sum_{i \in I} g_i$, $H \leftarrow \sum_{i \in I} h_i$
21: **for** m = 1 to M **do**
22: $G_L \leftarrow 0$, $H_L \leftarrow 0$
23: **for** j = 1 to K **do**
24: $G_L \leftarrow G_L + g_j$, $H_L \leftarrow H_L + h_j$
25: $G_R \leftarrow G - G_L$, $H_R \leftarrow H - H_L$
26: $Gain \leftarrow \max\left(Gain, \frac{G_L^2}{H_L + \lambda} + \frac{G_R^2}{H_R + \lambda} - \frac{G^2}{H + \lambda}\right)$
27: **end for**
28: **end for**
29: $\hat{Scor}_i \leftarrow \hat{Scor}_i + \hat{S}_{it}$

Output: $\hat{\mathbf{Scor}} = \left\{\hat{Scor}_1, \hat{Scor}_2, \cdots, \hat{Scor}_{Num}\right\}$,

able *Loss* by Eq. (20):

$$\begin{aligned} Loss\left(Scor_i, \hat{Scor}_i\right) &= Loss\left(Scor_i, \hat{Scor}_i^{(t-1)} + Fit_t(Scor_i)\right) \\ &\approx Loss\left(Scor_i, \hat{Scor}_i^{(t-1)}\right) + g_i Fit_t(Scor_i) \\ &\quad + \frac{h_i}{2} Fit_t^2(Scor_i), \end{aligned} \tag{20}$$

where $g_i = \partial_{\hat{Scor}_i} Loss\left(Scor_i, \hat{Scor}_i\right)$ and $h_i = \partial^2_{\hat{Scor}_i} Loss\left(Scor_i, \hat{Scor}_i\right)$ respectively represent the first derivative and the second derivative of the loss function l. In addition, The second term of the Eq. (16) is the regular term to control the complexity of the model and prevent over-fitting, which can be represented as follows:

$$\Omega(f_t) = \gamma M + \frac{1}{2}\lambda \sum_{m=1}^{M} w_m^2, \tag{21}$$

where γ and λ are the key hyper-parameters to control the model complexity. M means the number of leaf nodes and $\frac{1}{2}\sum_{m=1}^{M} w_m^2$ represents the L2 norm of leaf nodes. To complete node split by traversing all features with greedy algorithm, Eq. (16) is transformed into a new format as follows:

$$Obj^* = -\frac{1}{2} \sum_{j=1}^{M} \frac{(\sum_{i \in I_j} g_i)^2}{\frac{1}{2}\sum_{i \in I_j} h_i + \lambda} + \gamma M. \tag{22}$$

Theoretically, the smaller value of the objective function (Eq. (22)) represents the better tree structure we can retrieve. When the value of Obj^* is the minimum, we can achieve the optimal solution. To find the optimal tree structure, the exact greedy algorithm is used to divide the existing leaf nodes, which is based on the maximum difference of Obj^* values between before-split and after-split trials. A modified formula for calculating *Gain* is as follows:

$$Gain = \frac{1}{2}\left[\frac{\left(\sum_{i \in I_L} g_i\right)^2}{\sum_{i \in I_L} h_i + \lambda} + \frac{\left(\sum_{i \in I_R} g_i\right)^2}{\sum_{i \in I_R} h_i + \lambda} - \frac{(\sum_{i \in I} g_i)^2}{\sum_{i \in I} h_i + \lambda}\right] - \gamma, \tag{23}$$

where I_L, I_R and I are sets of samples assigned to the left and right nodes after split and overall samples before split respectively. In Eq. (23), the first, second, and third terms are the scores of left subtree, right subtree, and total subtree, respectively. The parameter γ is the extra overhead introduced by adding a new leaf node.

The right part of Fig. 8 shows the design of our gradient tree ensemble model. First, the sorted academic features (Taking the prediction of Probability Statistics as an example, Table 1 illustrates the features from 2 dimensions to 16 dimensions.) are obtained from the memory and the corresponding gradient statistics are acquired from the high-dimensional kernel space. Then, such features are traversed through the greedy algorithm and the weight w on each leaf node is calculated by the gradient statistics as the predicted value $\hat{S}_{it}$ of corresponding samples. The optimal split of each leaf node is found according to Obj^*. In this process, the regular item Ω controls the complexity of trees, and its parameters λ and γ are valued by the grid search algorithm. Finally, as shown in the bottom of Fig. 8, the model casts up the weights of correlated leaf nodes as predicted values for the ith sample. For example, the predicted score $\hat{Scor}_1$ of the first student $Samp_1$ from **Tree 1** (CGPA $\rightarrow$ Scholarship $\rightarrow$ BT $\rightarrow w_1$) and **Tree 2** (AS $\rightarrow$ BF $\rightarrow w_6$) is the summation of $w_1 + w_6$.

Implementation : We implement XGBoost-enhanced SAP prediction with scalable metadata clustering in ProbSAP with dataset $D = \{Samp_1, Samp_2, \cdots, Samp_{Num}\}$ containing *Num* instances as displayed in Algorithm 1. In order to alleviate the imbalance of academic data, we first set up a scalable metadata clustering structure to assign categories for each sample in Lines 2–16. Specifically, Lines 2–5 select the SSE curve turning point to configure the initial cluster centroid number K and Lines 6–7 update the K value by calculating the point average distortion AD_k and difference D_k. After properly selecting K, Lines 8–9 implement the selection of initial cluster centroids by constructing circles with center $\bar{\mu}$ and radius r, and the clustering process based on k-means is completed in Lines 10–16. Then XGBoost-enhanced SAP prediction module continues to categorize the output of data clustering. Lines 18–20 initialize the tree's split gain *Gain*, predicted score $\hat{Scor}_i$, and gradient statistics G and H. Lines 21–29 iteratively build prediction models based on the greedy algorithm. Finally, Line 21 generates score prediction results $\hat{\mathbf{Scor}} = \left\{\hat{Scor}_1, \hat{Scor}_2, \cdots, \hat{Scor}_{Num}\right\}$.

5. Experimental evaluation

5.1. Experimental setup

Configuration: Our experiments are conducted in Windows 10 operating system, consisting of two 10-core Intel Xeon 4210R processors. And our computing node is equipped with 32GB DRAM

and 2TB SATA. NVIDIA Geforce GTX 1080ti can support multiple CUDA cores. In the experiment, we implement ProbSAP with MATLAB 2020b and launch it on multiple threads.

Metrics: To assess ProbSAP's potential to support final mark prediction, we employ eight metrics to evaluate the models from three aspects. They are standard metrics for evaluating regression problems in this study that predict continuous output values such as final exam scores. Firstly, mean absolute error (MAE), root mean squared error (RMSE), mean absolute percentage error (MAPE) are used to analyze the accuracy of the predicted results as follows:

$$\begin{aligned} \mathrm{MAE} &= \frac{1}{Num}\sum_{i=1}^{Num}\left|Scor_i - \hat{Scor}_i\right|, \\ \mathrm{RMSE} &= \sqrt{\frac{1}{Num}\sum_{i=1}^{Num}\left(Scor_i - \hat{Scor}_i\right)^2}, \\ \mathrm{MAPE} &= \frac{100\%}{Num}\sum_{i=1}^{Num}\left|\frac{Scor_i - \hat{Scor}_i}{Scor_i}\right|, \end{aligned} \tag{24}$$

where Num is the number of samples, $Scor_i$ and $\hat{Scor}_i$ respectively represent the actual score of the ith sample and the predicted final marks by the model. These three statistical criteria can evaluate how close the predicted results are to the real data. A smaller value means a better fit and zero means a perfect fitting. And then correlation coefficient (C) and coefficient of determination (R^2) are also used to measure how well predicted values explain actual scores and how closely the two results were related by Eq. (25):

$$\mathrm{C} = \frac{\mathrm{Cov}(Scor_i, \hat{Scor}_i)}{\sqrt{\mathrm{Var}(Scor_i)\,\mathrm{Var}(\hat{Scor}_i)}},\quad \mathrm{R}^2 = 1 - \frac{\sum_{i=1}^{Num}\left(Scor_i - \hat{Scor}_i\right)^2}{\sum_{i=1}^{Num}\left(Scor_i - \overline{Scor}_i\right)^2}, \tag{25}$$

where $\mathrm{Cov}(Scor_i, \hat{Scor}_i)$ stands for the covariance of $Scor_i$ and $\hat{Scor}_i$; $\mathrm{Var}(\mathrm{Scor_i})$ and $\mathrm{Var}(\hat{\mathrm{Scor}}_i)$ calculate the variance; $\overline{Scor}_i$ is the average actual score. The values of C and R^2 closer to 1 mean that the predicted values can explain more about actual scores.

To further investigate the credibility of ProbSAP, Nash-Sutcliffe efficiency coefficient (NASH), variance account for (VAF), and a10-index are used in our evaluation, which could be computed by the following equations respectively:

$$\begin{aligned} \mathrm{NASH} &= 1 - \left[\frac{\frac{1}{Num}\sum_{i=1}^{Num}\left(Scor_i - \hat{Scor}_i\right)^2}{\frac{1}{Num}\sum_{i=1}^{Num}\left(\hat{Scor}_i - Scor_{i,avr}\right)^2}\right], \\ \mathrm{VAF} &= \left[1 - \frac{\mathrm{Var}\left(Scor_i - \hat{Scor}_i\right)}{\mathrm{Var}\left(Scor_i\right)}\right] \times 100, a10-\mathrm{index} = \frac{Num_{10}}{Num}, \end{aligned} \tag{26}$$

where $Scor_{i,avr}$ is the average actual score of all samples. Num_{10} is the number of samples with an error of less than 10% between actual value and predicted value, that is, $0.9 < Scor_i/\hat{Scor}_i < 1.1$. The model performs well with NASH and a10-index close to 1, or VAF close to 100. In our experiment, **training set** means the cross-validated instance set for model construction and adjustment. and **testing set** means the varying instance set for model evaluation, accounting for 80% and 20% of the overall set respectively.

5.2. Model verification and evaluation

1) Overall performance for different methods: Table 2 summarizes the eight evaluation metrics and compares the academic performance prediction results of the five methods (CNN, SVR, RFR, XGBoost, Catboost-SHAP [19], AS-SAN [20], and our new ProbSAP). Specifically, for training set, Catboost-SHAP and AS-SAN achieve smaller MAE, RMSE, and MAPE than RFR and XGBoost, but ProbSAP outperforms them by having lower outcomes (1.3728, 1.8584, 0.0256). And ProbSAP delivers outstanding performance at the correlation degree of C and R^2 such as the better correlation results closer to 1 compared to CNN, SVR, RFR, and XGBoost. In addition, ProbSAP benefits from collaborative data processing and scalable metadata clustering. Hence, its NASH, VAF and a10-index are significantly higher than other cases. To evaluate the benefit, we compare MAE, RMSE, MAPE, C, R^2, NASH, VAF, and a10-index of ProbSAP with those of CNN, SVR, RFR, XGBoost, Catboost-SHAP, and AS-SAN in the testing set. ProbSAP still outperforms other methods by minimizing the inter-class imbalance with partitional clustering. As we can see, ProbSAP yields the best performance for both training set and testing set.

To further evaluate the performance of ProbSAP, we visualize the absolute error (AE) of the actual and predicted values and compare the performance of ProbSAP with CNN, SVR, RFR, XGBoost, Catboost-SHAP, and AS-SAN in Fig. 10. The AE visualization of ProbSAP is illustrated in the last column. In this experiment, we randomly selected 250 data points from 4420 samples, and divided the absolute error interval into [0, 1), [1,3), [3,6), [6,10) and [10,100] to keep the sample distribution balanced. In Fig. 10, it can be observed that our ProbSAP achieves the best performance for both training and testing sets because collaborative data processing and clustering make the largest number of samples close to the perfect fitting line. RFR and XGBoost also make more samples concentrate on the perfect fitting line compared to CNN and SVR for training sets. But RFR performs better on testing set than CNN, SVR, and XGBoost. The latest EDM methods such as Catboost-SHAP and AS-SAN achieve better results with smaller errors than other standard methods, but our ProbSAP still outperforms the latest EDM methods for various datasets. Furthermore, compared to the poor generalization of XGBoost due to the data imbalance issue, ProbSAP greatly benefits the imbalanced class prediction with more concentrated data distribution.

2) Performance impact of different K values and features: Table 3 compares the results of eight evaluation metrics of Prob-

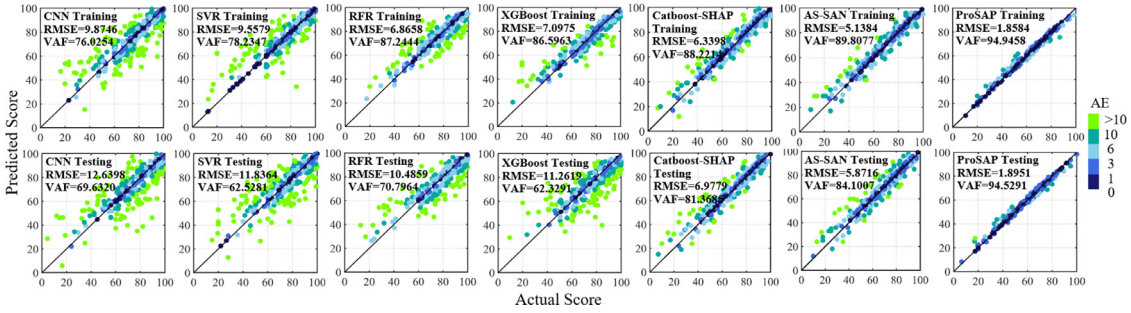


Fig. 10. Comparison graph of AE visualization of seven different methods for the training and testing sets. In the figure, the black line represents the perfect fitting line and the points of different colors represent different AE intervals.

Table 2
Comparison of prediction performance of ProbSAP with CNN, SVR, RFR, XGBoost, Catboost-SHAP, and AS-SAN using eight evaluation metrics.

	Training							
Method	MAE	RMSE	MAPE	C	R^2	NASH	VAF	a10-index
CNN	7.7335	9.8746	0.1577	0.8180	0.7027	0.5530	76.0254	0.6547
SVR	6.6414	9.5579	0.1260	0.8648	0.7479	0.6498	78.2347	0.6800
RFR	5.0906	6.8658	0.1031	0.9411	0.8821	0.8142	87.2444	0.7249
XGBoost	5.0467	6.5975	0.0870	0.9420	0.8880	0.8440	86.7963	0.7395
Catboost-SHAP	4.2631	6.3398	0.0725	0.9590	0.9107	0.9065	88.2214	0.8501
AS-SAN	3.9857	5.1384	0.0608	0.9620	0.9280	0.9157	89.8077	0.8923
ProbSAP(K=4)	**1.3728**	**1.8584**	**0.0256**	**0.9904**	**0.9821**	**0.9792**	**94.9458**	**0.9913**
	Testing							
Method	MAE	RMSE	MAPE	C	R^2	NASH	VAF	a10-index
CNN	9.5665	12.6398	0.1762	0.7765	0.6029	0.4504	59.6320	0.5279
SVR	8.5740	11.8364	0.1623	0.7856	0.6271	0.5209	62.5281	0.5725
RFR	8.8146	10.4859	0.1518	0.7863	0.6182	0.5569	70.7964	0.5200
XGBoost	9.3280	11.2619	0.1671	0.7563	0.6061	0.5282	62.3291	0.5050
Catboost-SHAP	5.0989	6.9779	0.0905	0.9155	0.8494	0.7647	81.3682	0.7841
AS-SAN	4.2431	5.8716	0.0712	0.9318	0.8813	0.8028	84.1007	0.8170
ProbSAP(K=4)	**1.4219**	**1.8951**	**0.0264**	**0.9856**	**0.9807**	**0.9782**	**94.5291**	**0.9866**

Table 3
Comparison of prediction performance of ProbSAP with AS&BF features and K=4 under different categories using eight evaluation metrics.

	Training							
Category	MAE	RMSE	MAPE	C	R^2	NASH	VAF	a10-index
C1	1.3888	1.9248	0.0164	0.9836	0.9674	0.9593	96.4677	0.9954
C2	2.4676	3.3852	0.0376	0.9766	0.9538	0.9359	94.7996	0.9402
C3	0.7257	1.0581	0.0128	0.9985	0.9969	0.9964	99.6544	0.9798
C4	1.5386	2.1853	0.0305	0.9923	0.9847	0.9813	98.3048	0.9608
Overall	**1.3728**	**1.8584**	**0.0256**	**0.9904**	**0.9821**	**0.9792**	**94.9458**	**0.9913**
	Testing							
Category	MAE	RMSE	MAPE	C	R^2	NASH	VAF	a10-index
C1	1.5096	1.9760	0.0176	0.9809	0.9622	0.9482	95.7131	0.9954
C2	2.9498	4.0351	0.0444	0.9617	0.9248	0.9002	92.0204	0.9080
C3	0.8490	1.3665	0.0208	0.9981	0.9962	0.9957	99.5915	0.9798
C4	1.5524	2.2618	0.0318	0.9916	0.9833	0.9796	98.1601	0.9500
Overall	**1.4219**	**1.8951**	**0.0264**	**0.9856**	**0.9807**	**0.9782**	**94.5291**	**0.9866**

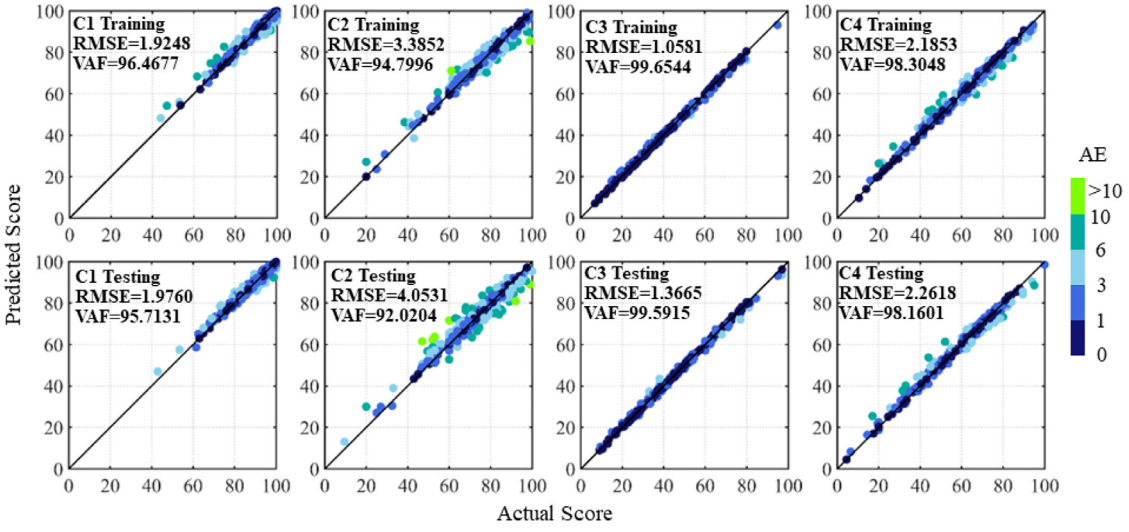
SAP with AS&BF features under different categories when K=4. In this case, instances are clustered into 4 categories (C1, C2, C3, and C4) based on the original distribution of AS&BF features to conduct XGBoost-enhanced SAP prediction and each category shows similar performance because the data points are distributed in a balanced manner as shown in Fig. 6. In detail, C1, C2, C3, and C4 represent the score ranges of 60–100, 40–100, 10–80, and 0–90 respectively. ProbSAP performs well on testing set as well as on training set from C1 to C4 due to its clustering mechanism, which illustrates the superior robustness of ProbSAP framework. Clearly, ProbSAP achieves the best performance in Category C3 for both training and testing sets. To explore the performance impact of clustering features, we further choose CGPA&BF features to evaluate scalable metadata clustering of ProbSAP. In the experiment, similar prediction results of eight evaluation metrics are shown in Table 4. ProbSAP improves the prediction performance by making full use of training data with CGPA&BF features for the training set, benefiting the alleviation of information loss. But it is not effective to track the student overall performance based on totally different testing set with higher errors (MAE: 3.2139, RMSE: 6.5722, MAPE: 0.0881) and lower efficiency values (NASH: 0.7845, VAF: 90.0478, a10-index: 0.9127), indicating the poor generalization of CGPA&BF feature-based prediction.

Fig. 11(a) and (b) present the AE visualization of ProbSAP under different categories (C1, C2, C3, and C4) when K=4. Those four categories focus on the different segments, especially the scores of samples in C1 are generally high, which coincide with the centroid positions of BF, AS and CGPA in Fig. 6. In addition, ProbSAP obtains poor prediction outcomes for C2, C3 and C4 with CGPA&BF features because CGPA cannot capture students' continuous academic performance. However, the instances are compactly distributed on both sides of the black perfect fitting line in training and testing sets. In a word, the overall performance of ProbSAP is impacted by clustering features, but it is superior and stable because its scalable clustering algorithm can separate various students to alleviate academic data imbalance.

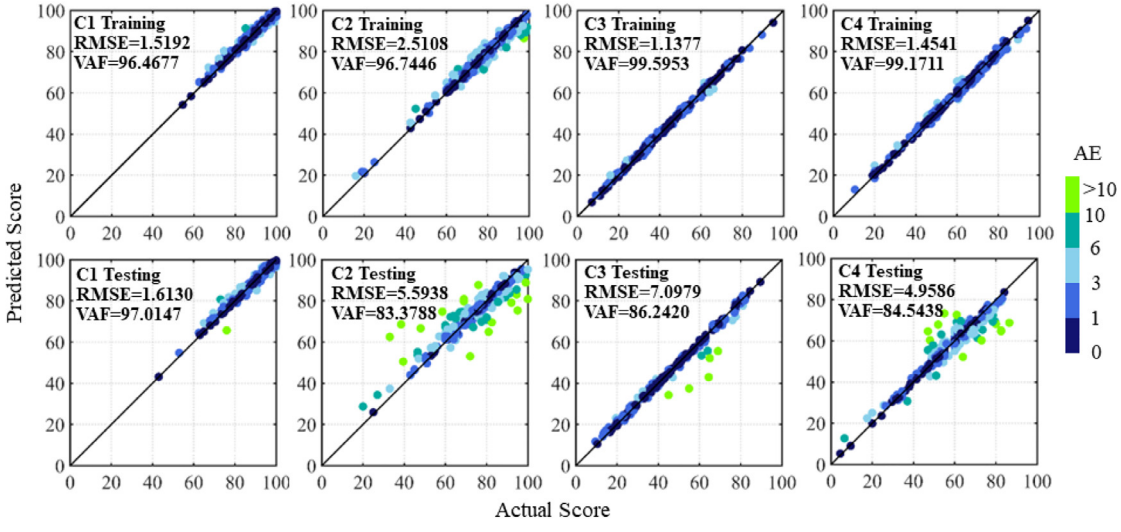
Table 5 compares the results of eight evaluation metrics of ProbSAP with AS&BF features under five different categories (C1, C2, C3, C4, and C5) when K=5. In detail, each category from C1 to C5 shows stable results and the ProbSAP framework achieves more comprehensive performance than the case of K=4 due to the average distance between inter-class instances has been reduced by fine-grained sample division. Then we further set CGPA&BF features to evaluate scalable metadata clustering of ProbSAP in Table 6. With the increasing number of categories in training set, ProbSAP significantly reduces prediction errors such as MAE, RMSE, and MAPE and dramatically preserves credibility by NASH, VAF, and a10-index. Fig. 12(a) and (b) show the AE visualization of ProbSAP with different features when K=5. The prediction results with K=5 are closer to actual scores than the case with K=4 in

Table 4
Comparison of prediction performance of ProbSAP with CGPA&BF features and K=4 under different categories using eight evaluation metrics.

	Training							
Category	MAE	RMSE	MAPE	C	R^2	NASH	VAF	a10-index
C1	1.0673	1.5192	0.0123	0.9875	0.9752	0.9700	97.3273	1.0000
C2	1.8086	2.5108	0.0265	0.9853	0.9708	0.9622	96.7446	0.9749
C3	0.7919	1.1377	0.0218	0.9982	0.9963	0.9958	99.5953	0.9826
C4	1.0611	1.4541	0.0219	0.9963	0.9926	0.9911	99.1711	0.9850
Overall	**1.2014**	**1.5697**	**0.0175**	**0.9863**	**0.9699**	**0.9658**	**96.8967**	**0.9924**
	Testing							
Category	MAE	RMSE	MAPE	C	R^2	NASH	VAF	a10-index
C1	1.0429	1.6130	0.0122	0.9855	0.9713	0.9671	97.0147	0.9962
C2	3.4656	5.5938	0.0533	0.9177	0.8423	0.7559	83.3778	0.8807
C3	2.7160	7.0979	0.0493	0.9300	0.8649	0.8131	86.2420	0.8762
C4	3.2246	4.9586	0.0483	0.9231	0.8564	0.7750	84.5438	0.8917
Overall	**3.2139**	**6.5722**	**0.0881**	**0.9255**	**0.9052**	**0.7845**	**90.0478**	**0.9127**



(a) ProbSAP with AS&BF features.



(b) ProbSAP with CGPA&BF features.

Fig. 11. Comparison of AE visualization of ProbSAP framework with different features when the number of categories K is 4. In the figure, C1, C2, C3 and C4 represent four different categories, the black line represents the perfect fitting line and the points of different colors represent different AE intervals.

Table 5
Comparison of prediction performance of ProbSAP with AS&BF features and K=5 under different categories using eight evaluation metrics.

	Training							
Category	MAE	RMSE	MAPE	C	R^2	NASH	VAF	a10-index
C1	1.9782	2.7528	0.0293	0.9835	0.9673	0.9577	96.3833	0.9667
C2	1.1867	1.6470	0.0139	0.9877	0.9755	0.9705	97.3688	0.9992
C3	0.8999	1.2561	0.0149	0.9970	0.9941	0.9931	99.3489	0.9950
C4	1.0837	1.5418	0.0246	0.9966	0.9931	0.9919	99.2367	0.9670
C5	0.3508	0.5145	0.0092	0.9996	0.9992	0.9991	99.9152	1.0000
Overall	**1.1023**	**1.5728**	**0.1148**	**0.9898**	**0.9756**	**0.9811**	**98.0152**	**0.9853**
	Testing							
Category	MAE	RMSE	MAPE	C	R^2	NASH	VAF	a10-index
C1	2.0266	2.8782	0.0303	0.9797	0.9598	0.9490	95.6725	0.9600
C2	1.1782	2.3104	0.0210	0.9723	0.9454	0.9263	94.0219	0.9924
C3	0.9404	2.2662	0.0146	0.9889	0.9779	0.9765	97.7709	0.9940
C4	1.2589	1.8722	0.0312	0.9948	0.9895	0.9876	98.8721	0.9554
C5	0.4342	0.6031	0.0116	0.9995	0.9991	0.9988	99.8848	1.0000
Overall	**1.2564**	**1.7865**	**0.0187**	**0.9821**	**0.9688**	**0.9614**	**96.2317**	**0.9658**

Table 6
Comparison of prediction performance of ProbSAP with CGPA&BF features and K=5 under different categories using eight evaluation metrics.

	Training							
Category	MAE	RMSE	MAPE	C	R^2	NASH	VAF	a10-index
C1	2.1838	2.9612	0.0311	0.9779	0.9563	0.9418	95.1747	0.9635
C2	1.1601	1.6198	0.0132	0.9856	0.9714	0.9654	96.9398	1.0000
C3	1.2905	1.7811	0.0223	0.9942	0.9884	0.9852	98.6630	0.9839
C4	0.6997	1.0066	0.0153	0.9984	0.9969	0.9964	99.6553	0.9903
C5	0.4208	0.6188	0.0106	0.9994	0.9989	0.9988	99.8818	0.9945
Overall	**1.1896**	**1.7532**	**0.0145**	**0.9874**	**0.9815**	**0.9726**	**97.8835**	**0.9912**
	Testing							
Category	MAE	RMSE	MAPE	C	R^2	NASH	VAF	a10-index
C1	2.3178	3.2400	0.0327	0.9721	0.9450	0.9289	94.1101	0.9639
C2	1.1583	1.5637	0.0131	0.9857	0.9716	0.9682	97.0855	1.0000
C3	1.5857	2.5562	0.0274	0.9849	0.9701	0.9623	96.7266	0.9706
C4	2.0342	3.0535	0.0529	0.9849	0.9700	0.9627	96.7508	0.8846
C5	0.5824	0.7707	0.0155	0.9991	0.9982	0.9980	99.8031	0.9886
Overall	**1.4637**	**1.9685**	**0.0269**	**0.9815**	**0.9750**	**0.9703**	**96.9568**	**0.9757**

Fig. 11(a) and (b) and the prediction performance of ProbSAP with CGPA&BF features is almost the same as AS&BF features, which visually demonstrates that ProbSAP benefits the high robustness prediction with more precise categorization.

Tables 7 and 8 separately summarize the evaluation metrics of six categories (C1, C2, C3, C4, C5, and C6) of ProbSAP with AS&BF and CGPA&BF features when the number of categories K is 6. For both training and testing sets, ProbSAP always achieves excellent performance for different evaluation metrics (i.e., more than half of MAEs from C1 to C6 are less than 1 and all the a10-indexes are more than 0.9), and its experimental results are more stable than the cases of K=4 and K=5. Fig. 13(a) and (b) show AE visualization results with K=6. We can see that the performances of ProbSAP with AS&BF and CGPA&BF are comparable and all the sample points are closely distributed near the perfect fitting lines, which means that increasing K helps to divide samples more accurately to overcome the shortcoming of inter-class imbalance. Profiting from the fusion mechanism of scalable metadata clustering and improved XGBoost algorithm, ProbSAP categorizes samples and conducts in-depth mining of the real-world multi-source academic data to alleviate the imbalance and preserve academic performance with high robustness. As a consequence, our ProbSAP framework shows superiority of tracking student's multiple characteristics for academic performance prediction.

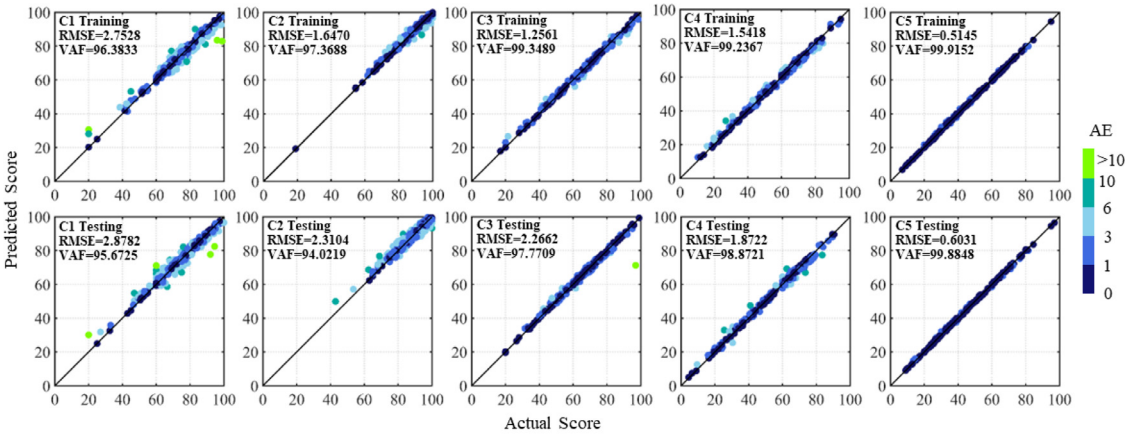
3) Feature importance estimation in ProbSAP: To further reveal the importance of different features for course final mark prediction, we select the following three indicators to evaluate the 15 features from ACS to ASCC in Table 1. **(1) Gain**, as a generalization concept of information gain, represents the contribution of each feature that causes the objective function to achieve a smaller value when a node splits. The relative contribution of a feature to the model is usually calculated by obtaining its contribution to all the trees in ProbSAP as follows:

$$Gain_{F_i} = \frac{\sum_{k=1}^{T}\sum_{j=1}^{N} Gain_{F_{ijk}}}{\sum_{m=1}^{M}\left(\sum_{k=1}^{T}\sum_{j=1}^{N} Gain_{F_{mjk}}\right)}, \tag{27}$$

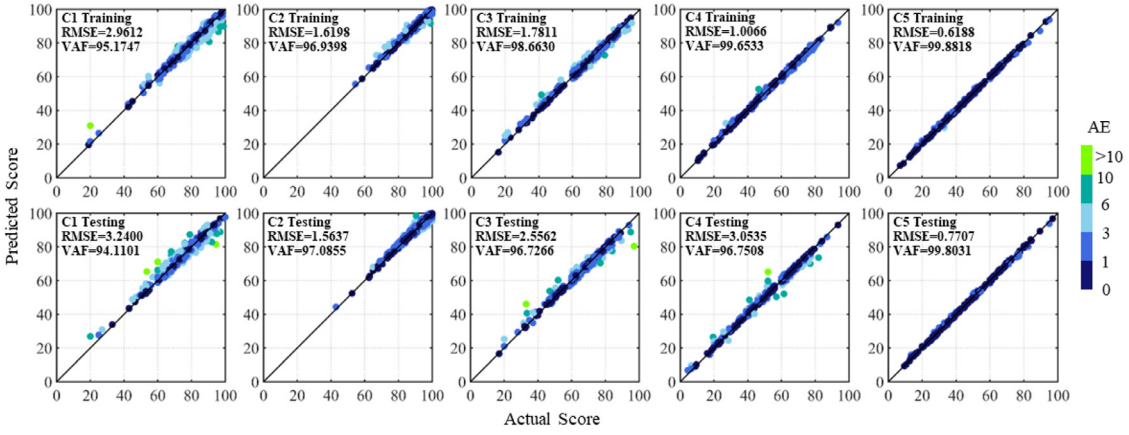
where N, T, and M are the numbers of split nodes in a tree, trees, and features respectively. **(2) Weight** represents the times a feature is treated as a split feature when the tree splits in XGBoost-enhanced modeling. The calculation of weight is the number of splits of a feature percentage of all features split as follows:

$$Weight_{F_i} = \frac{\sum_{k=1}^{T}\sum_{j=1}^{N} Times_{F_{ijk}}}{\sum_{m=1}^{M}\left(\sum_{k=1}^{T}\sum_{j=1}^{N} Times_{F_{mjk}}\right)}. \tag{28}$$

(3) Cover represents the relative number of samples associated with a feature, i.e., when the tree model splits, the number of samples covered by the feature that is a split node is divided by the



(a) ProbSAP with AS&BF features.



(b) ProbSAP with CGPA&BF features.

Fig. 12. Comparison of AE visualization of the ProbSAP framework with different features when the number of categories K is 5. In the figure, C1, C2, C3, C4 and C5 represent five different categories, the black line represents the perfect fitting line and the points of different colors represent different AE intervals.

Table 7
Comparison of prediction performance of ProbSAP with AS&BF features and K=6 under different categories using eight evaluation metrics.

	Training							
Category	MAE	RMSE	MAPE	C	R^2	NASH	VAF	a10-index
C1	1.7559	2.4199	0.0258	0.9873	0.9748	0.9654	97.0573	0.9768
C2	0.9584	1.3656	0.0110	0.9898	0.9796	0.9749	97.7578	0.9991
C3	2.0591	2.7354	0.0297	0.9832	0.9667	0.9576	96.3551	0.9704
C4	0.9633	1.3322	0.0214	0.9974	0.9948	0.9939	99.4239	0.9807
C5	0.4080	0.5873	0.0109	0.9995	0.9990	0.9989	99.8903	1.0000
C6	0.9617	1.3440	0.0177	0.9970	0.9940	0.9929	99.3281	0.9900
Overall	**0.9840**	**1.4822**	**0.0196**	**0.9923**	**0.9847**	**0.9807**	**99.1175**	**0.9859**
	Testing							
Category	MAE	RMSE	MAPE	C	R^2	NASH	VAF	a10-index
C1	1.8819	2.6545	0.0270	0.9844	0.9690	0.9556	96.3308	0.9781
C2	0.9978	1.4328	0.0112	0.9877	0.9756	0.9711	97.4071	1.0000
C3	1.9159	2.8414	0.0292	0.9803	0.9611	0.9507	95.7789	0.9663
C4	1.0007	1.3831	0.0224	0.9971	0.9942	0.9930	99.3458	0.9732
C5	0.4107	0.5600	0.0119	0.9995	0.9989	0.9989	99.8888	1.0000
C6	0.9318	1.3316	0.0168	0.9966	0.9932	0.9927	99.2991	0.9860
Overall	**0.9921**	**1.3564**	**0.0210**	**0.9914**	**0.9852**	**0.9819**	**99.0785**	**0.9867**

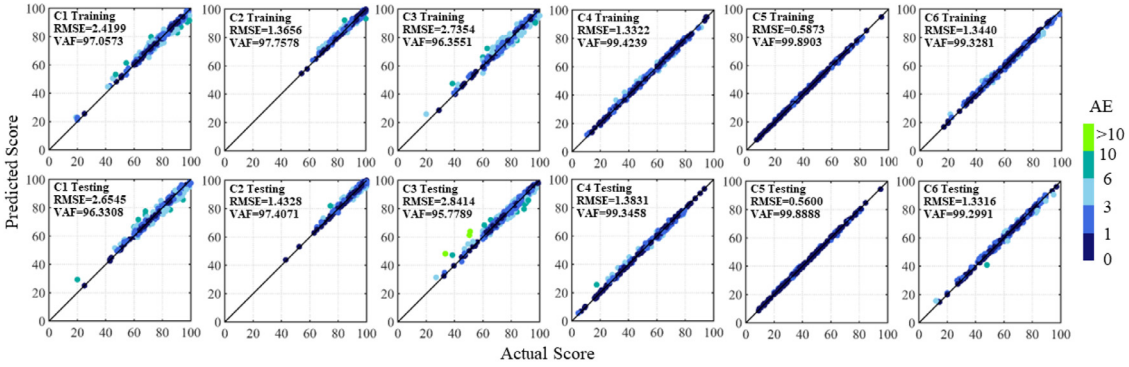
total number of samples as follows:

$$Cover_{F_i} = \frac{\sum_{k=1}^{T}\sum_{j=1}^{N} Num_{F_{ijk}}}{\sum_{m=1}^{M}\left(\sum_{k=1}^{T}\sum_{j=1}^{N} Num_{F_{mjk}}\right)}. \quad (29)$$

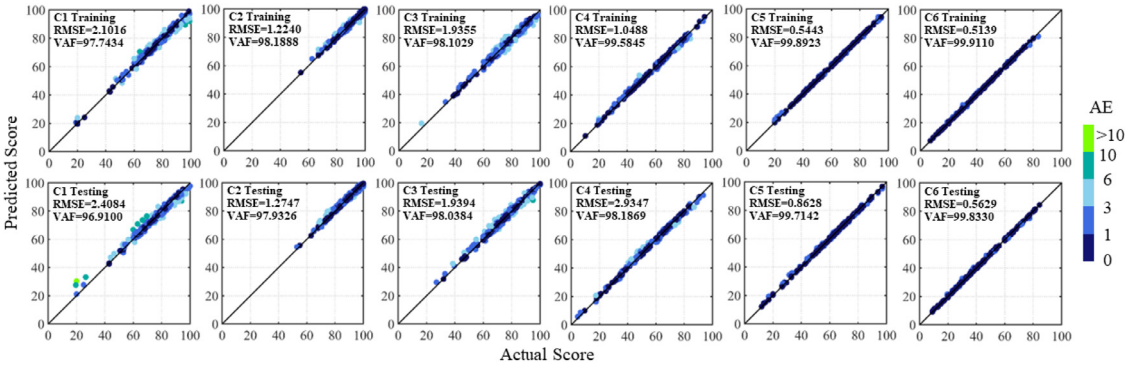
Fig. 14(a)–(c) show the comparison of feature importance under ProbSAP framework based on the gain, weight, and cover indicators. In detail, during the model building process, the top five features such as AS, CGPA, GPA_3, GPA_2, and BF contribute the most to the reduction of objective function values in Fig. 14(a). The five features including CGPA, AS, GPA_3, BF, and GPA_2 in Fig. 14(b) are the most frequently used as elemental features of each tree nodes. In Fig. 14(c), AS, CGPA, GPA_3, GPA_1, and BF cover the rela-

Table 8
Comparison of prediction performance of ProbSAP with CGPA&BF features and K=6 under different categories using eight evaluation metrics.

	Training							
Category	MAE	RMSE	MAPE	C	R^2	NASH	VAF	a10-index
C1	1.5140	2.1016	0.0223	0.9902	0.9804	0.9742	97.7434	0.9894
C2	0.8626	1.2240	0.0100	0.9917	0.9835	0.9800	98.1888	1.0000
C3	1.4435	1.9355	0.0211	0.9915	0.9830	0.9789	98.1029	0.9889
C4	0.7318	1.0488	0.0148	0.9981	0.9963	0.9956	99.5845	0.9941
C5	0.3892	0.5443	0.0075	0.9995	0.9991	0.9989	99.8932	1.0000
C6	0.3327	0.5139	0.0095	0.9996	0.9991	0.9991	99.9110	1.0000
Overall	**0.9795**	**1.3629**	**0.0192**	**0.9931**	**0.9860**	**0.9822**	**99.1089**	**0.9859**
	Testing							
Category	MAE	RMSE	MAPE	C	R^2	NASH	VAF	a10-index
C1	1.7409	2.4084	0.0260	0.9869	0.9740	0.9628	96.9100	0.9811
C2	0.9373	1.2747	0.0108	0.9906	0.9813	0.9768	97.9326	1.0000
C3	1.4993	1.9394	0.0218	0.9912	0.9825	0.9780	98.0384	1.0000
C4	2.1066	2.9347	0.0299	0.9778	0.9561	0.9420	98.1869	0.9567
C5	0.6439	0.8628	0.0129	0.9989	0.9978	0.9970	99.7142	0.9923
C6	0.3689	0.5629	0.0103	0.9995	0.9989	0.9988	99.8830	1.0000
Overall	**0.9914**	**1.3787**	**0.0218**	**0.9920**	**0.9841**	**0.9806**	**99.0525**	**0.9848**

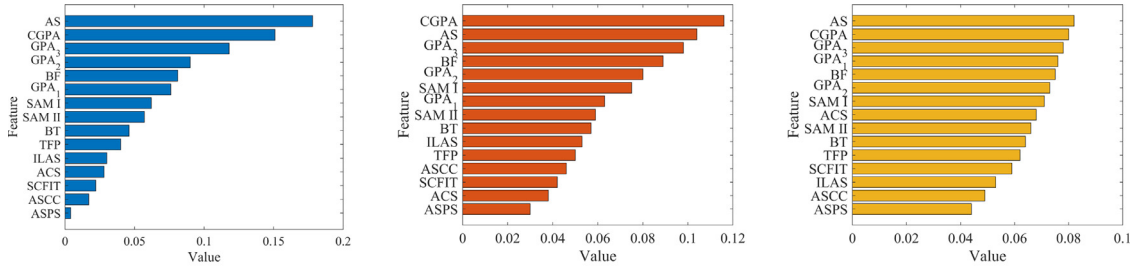


(a) ProbSAP with AS&BF features.



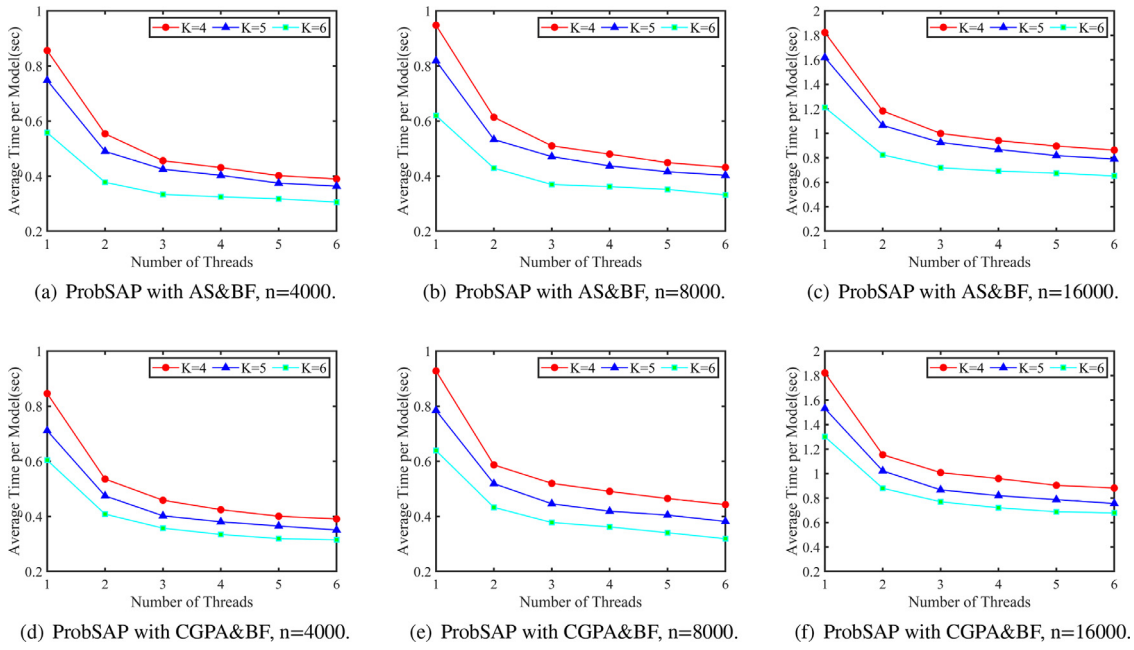
(b) ProbSAP with CGPA&BF features.

Fig. 13. Comparison of AE visualization of the ProbSAP framework with different features when the number of categories K is 6. In the figure, C1, C2, C3, C4, C5 and C6 represent six different categories, the black line represents the perfect fitting line and the points of different colors represent different AE intervals.



(a) Gain. (b) Weight. (c) Cover.

Fig. 14. Comparison of feature importance with three indicators under ProbSAP framework.



(a) ProbSAP with AS&BF, n=4000. (b) ProbSAP with AS&BF, n=8000. (c) ProbSAP with AS&BF, n=16000.

(d) ProbSAP with CGPA&BF, n=4000. (e) ProbSAP with CGPA&BF, n=8000. (f) ProbSAP with CGPA&BF, n=16000.

Fig. 15. Comparison of average time to train prediction models along with the number of threads for different clustering features and data sizes.

tively largest number of samples. However, it is obvious that the three features such as ASCC, ASPS, ACS, and ILAS are unnecessary to predict Probability Statistics score due to weak connection between different courses with the same attributes. Therefore, in consideration of student academic characteristics, the most important factors of affecting student performance are student's past academic performance (GPA-related features) and specific feature (BF). Due to the large differences in other math courses, the single course grade or teaching style cannot influence the student overall performance. Hence, we exploit the XGBoost-enhanced model to analyze the student's learning ability for academic performance prediction.

4) Efficiency analysis: Benefiting from the scalable metadata clustering and XGBoost-enhanced modeling module that support parallel processing at feature granularity, ProbSAP framework achieves better efficiency for student academic performance prediction. Before training, ProbSAP sorts out the features and launches their gradient statistics to high-dimensional kernel space for reuse, thereby reducing the amount of computation. Moreover, while building a tree model, the gain calculation of each feature can be performed simultaneously in multiple threads to reduce the modeling time. As we can see from Fig. 15(a)–(f), the average execution time of all cases drops along with the increasing thread number. For example, ProbSAP with AS&BF features and the number of samples n=4000, the average training time decreases from 0.84s by 1 thread to 0.32s by 6 threads for the case of K=4. Furthermore, to evaluate the computation overhead of ProbSAP with 6 threads, we compare the total execution time of ProbSAP with baseline XGBoost in Fig. 16. ProbSAP takes longer time than XGBoost while increasing the number of categories from K=4 to K=6. This is because of scalable metadata clustering that introduces additional computation. But the execution time just increases within 1s, which does not affect the superiority of our new ProbSAP framework. To concurrently predict multiple course marks according to the requirements of educational administration systems, ProbSAP has to be deployed in high-performance devices and support multithreading both in CPU and GPU modes in the future.

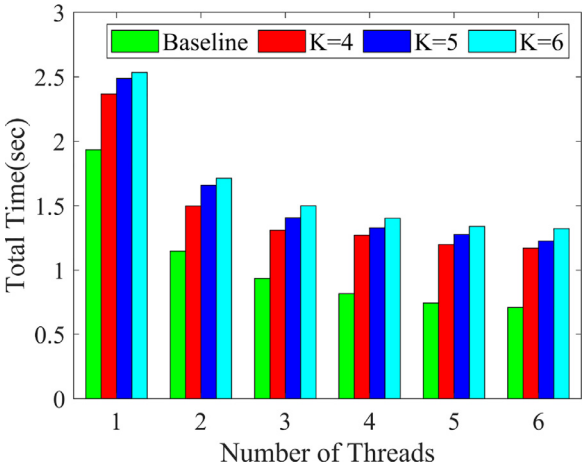
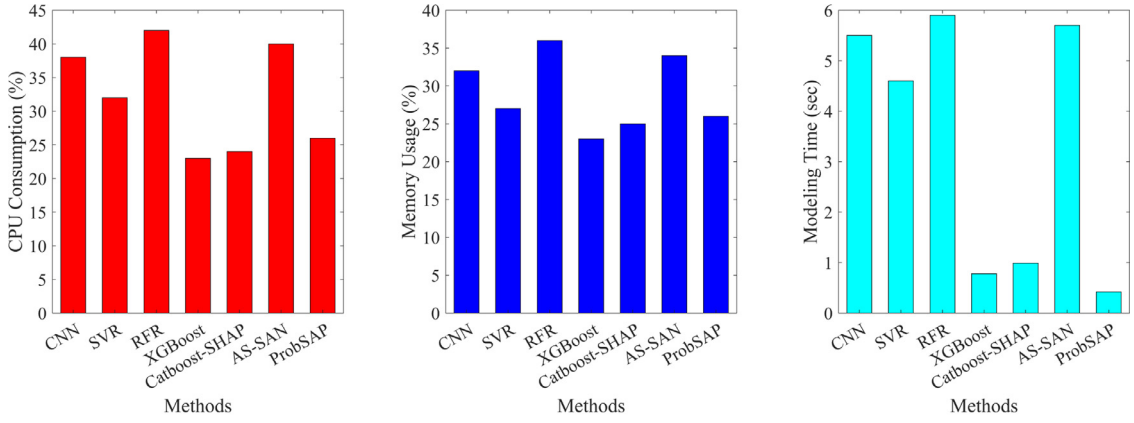


Fig. 16. Comparison of the total time spent on model training in different methods. In the figure, baseline represents baseline XGBoost method, K=4, 5, 6 represent the number of clustering categories under ProbSAP framework.

In addition, we evaluate the CPU consumption, memory usage, and modeling time of CNN, SVR, RFR, XGBoost, Catboost-SHAP, AS-SAN, and ProbSAP, as shown in Fig. 17(a)–(c). In Fig. 17(a), ProbSAP achieves a higher CPU consumption (26%) than XGBoost because of scalable metadata clustering. But compared to CNN (38%), SVR (32%), RFR (42%), and AS-SAN (40%), its consumption is much lower. For memory usage as shown in Fig. 17(b), ProbSAP outperforms CNN (32%), SVR (27%), RFR (36%), and AS-SAN (34%) by directly processing data in parallel threads. Due to the collaborative metadata structure, it delivers a slightly higher memory cost than XGBoost. However, the execution time of ProbSAP is significantly lower than other cases in Fig. 17(c). This finding demonstrates ProbSAP can guarantee the performance and efficiency in different cases while conducting prediction for college students.



(a) CPU consumption. (b) Memory usage. (c) Modeling time.

Fig. 17. Comparison of CPU consumption, memory usage, and modeling time of CNN, SVR, RFR, XGBoost, Catboost-SHAP, AS-SAN, and ProbSAP.

Table 9
Comparison of prediction performance of Catboost-SHAP, AS-SAN, and ProbSAP with SPD and OULAD datasets in [5] using eight evaluation metrics.

		Training							
Dataset	Method	MAE	RMSE	MAPE	C	R^2	NASH	VAF	a10-index
SPD	Catboost-SHAP	0.6389	0.7011	0.0079	0.9340	0.9282	0.9013	90.5297	0.9140
	AS-SAN	0.7914	0.8687	0.0092	0.9269	0.9018	0.8920	89.5312	0.8935
	ProbSAP	**0.3759**	**0.4581**	**0.0043**	**0.9752**	**0.9620**	**0.9487**	**94.8709**	**0.9634**
OULAD	Catboost-SHAP	3.9564	5.0130	0.0587	0.9598	0.9245	0.9046	89.9642	0.8929
	AS-SAN	3.8340	4.6259	0.0525	0.9613	0.9351	0.9289	90.2003	0.9043
	ProbSAP	**1.9728**	**2.6814**	**0.0397**	**0.9796**	**0.9520**	**0.9498**	**93.0129**	**0.9499**
		Testing							
Dataset	Method	MAE	RMSE	MAPE	C	R^2	NASH	VAF	a10-index
SPD	Catboost-SHAP	0.8537	0.9826	0.0092	0.9063	0.8992	0.8765	85.4661	0.8715
	AS-SAN	0.9183	1.0217	0.0099	0.8986	0.8814	0.8545	82.0273	0.8224
	ProbSAP	**0.5419**	**0.6564**	**0.0052**	**0.9411**	**0.9213**	**0.9145**	**91.0912**	**0.9130**
OULAD	Catboost-SHAP	4.7928	5.4187	0.0725	0.9117	0.8964	0.7910	82.8540	0.8795
	AS-SAN	4.4239	5.1560	0.0657	0.9248	0.8968	0.8816	85.4822	0.8857
	ProbSAP	**3.0344**	**3.9108**	**0.0551**	**0.9573**	**0.9318**	**0.9041**	**90.8556**	**0.9215**

5) Analysis of performance benchmarking: We use the standard educational dataset, student performance dataset (SPD) and open university learning analytics dataset (OULAD) in [5] to evaluate the performance of ProbSAP and validate our design through analysis of benchmark performance. Table 9 compares the prediction performance of Catbooos-SHAP, AS-SAN, and ProbSAP with eight evaluation metrics. For the SPD, ProbSAP delivers outstanding performance improvement at different evaluation metrics. For example, it reduces MAEs by 53% and 41% compared with AS-SAN and Catboost-SHAP in training set, respectively. In contrast, ProbSAP preserves the credibility by higher NASH, VAF, and a10-index in both training and testing sets. This is because of the additional benefits of collaborative data processing and scalable metadata clustering in ProbSAP. For the OULAD dataset, ProbSAP still outperforms Catboost-SHAP and AS-SAN by having lower prediction errors (MAE: 3.0344, RMSE: 3.9108, MAPE: 0.0551) and higher efficiency values (NASH: 0.9041, VAF: 90.8556, a10-index: 0.9215) in testing set. The validation results based on SPD and OULAD demonstrate that the fusion mechanism of scalable metadata clustering and XGBoost-enhanced SAP prediction can be extended to effectively track student's multiple characteristics from other educational datasets.

5.3. Summary

The aforementioned experimental results and evaluation demonstrate the superiority of ProbSAP in the prediction accuracy and efficiency. In this section, we will discuss its advantages, applicability, and limitations according to our experimental results.

1) Advantages and applicability of ProbSAP: ProbSAP achieves the best prediction accuracy for imbalanced data. The overall MAE of ProbSAP is less than 1.4219 for educational administration systems, which can meet the need of practical applications. It also shows strong reliability and generalization ability for different categories (C1, C2, C3, C4, C5, and C6). Its advantages of dealing with imbalanced data make it more suitable for predictive works with multi-dimensional data from educational settings. Additionally, ProbSAP reduces CPU consumption, memory usage, and modeling time. Our new ProbSAP framework, including collaborative data processing, scalable metadata clustering, and XGBoost-enhanced SAP prediction, can be used as a backbone by other researchers who perform tasks with imbalanced and multi-dimensional data samples.

2) Limitations of ProbSAP: On one hand, because ProbSAP provides the limited support to the single course final mark predic-

tion in CPU mode, ProbSAP has special requirements for more prediction tasks (i.e., predicting multiple course marks concurrently) to take full advantage of computation and memory resources. On the other hand, ProbSAP aims to explore many diverse and varied sources of offline educational information such as course information, students' behavior, students' performance, curriculum, etc., which is collected in traditional classroom or campus environment. But for distance education especially the most used web-based education, ProbSAP might not capture contextual information from a raw trace of students' navigation because the distance education system just provides a single log file for all the students with inherent data limitations such as simple access logs and similar online instruction records. Currently, we use a large quantity of daily activity data in collaborative data processing module to deeply probe student academic characteristics and integrate large amounts of activity profiles into EDM algorithms. For e-learning education systems, ProbSAP needs to be tuned to provide a more effective data processing module because heterogeneous student data such as online study time and assignment submission are gathered in relational databases. In online-offline blended teaching mode, the prediction performance of ProbSAP may be affected by the inconsistent and hysteretic information when asynchronous instructional feedback is perceived in online and offline learning. ProbSAP cannot deal with video stream data in the online classroom because streaming videos bring huge burden of data processing and storage.

6. Conclusions and future work

This study proposes a comprehensive and high-performance prediction framework, namely ProbSAP, which consists of the collaborative data processing module, the scalable metadata clustering framework, and the XGBoost-enhanced SAP prediction method. The collaborative data processing module can generate a new metadata matrix to alleviate educational data imbalance. Furthermore, the scalable metadata clustering framework divides multiple samples to different categories to further solve the data imbalance problem for multi-dimensional academic data. Finally, XGBoost-enhanced SAP prediction module with precise residual calculation and XGBoost-enhanced modeling implements course final mark prediction on the real-world data. Our validation results demonstrate that ProbSAP efficiently supports a variety of challenging imbalanced data and outperforms the state-of-the-art EDM methods such as Catboost-SHAP and AS-SAN. Particularly, ProbSAP can support multi-source student behavior information with the lowest predicted errors and the highest model credibility. ProbSAP also provides a reliable means for teachers to intervene with students in advance. Meanwhile, our results prove that in addition to previous grades, the frequency and time of breakfast are also closely related to students' course final mark, which provides guidance for higher education.

ProbSAP can be used by multiple applications within the similar prediction tasks such as the predictions of population growth, commodity demand, and stock price trends with multiple imbalanced data structures. These predictions are accompanied by salient features such as multi-dimensionality and imbalance, similar to the academic situation in this study. The collaborative data processing, scalable metadata clustering, and XGBoost-enhanced prediction can solve the above-mentioned practical problems as appropriate datasets are introduced. ProbSAP can also be utilized in classification problems such as fraud and earthquakes, when the loss function in XGBoost-enhanced method is set to cross entropy or hinge loss.

In the future, we continue to implement ProbSAP in GPU mode instead of CPU mode for obtaining parallel computation efficiency and perform more experiments on more educational datasets. Moreover, we plan to design an in-memory clustering module for efficient large-scale data organization. Hence a possible research direction for this study will involve designing in-memory structures to apply a continuous memory space for multiple features to achieve fast access and update. In consideration of student features associated with time, i.e., breakfast behavior, we can design a novel learning mechanism to systematically track the student behaviors from a temporal perspective.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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Appendix

Table A.1
Feature description of basic information and student behaviors from multi-source datasets for college students.

Data Category		Feature	Description	Amount	Creator
Student i	Student information	SID	Unique student identity	108,034	Student
		SNAME	Student's name		
		SGENDER	Student's gender		
		AGE	Student's age		
		SSID	Student's social security number		
		HOME	Student's home address		
		ETIME	Student's enrollment time		
		MNAME	Student's major name		
		MID	Student's major ID		
		STATUS	Student's school status		

(*continued on next page*)

Table A.1 (*continued*)

Data Category		Feature	Description	Amount	Creator
Behavior j	Course selection	SEMESTER	Semester of the course set up	775,500	Educational administration system
		CNAME	Course's name		
		SID	ID of student who has selected the course		
		DEPT	Department that offers the course		
		TID	Teacher identity who teaches the course		
		CTIME	Specific course start time		
		CREDIT	Course credit information		
	Exam performance	SNAME	Student name who takes the test	3,206,314	Teacher
		SID	Student ID who takes the test		
		CNAME	Testing course name		
		SEMESTER	Semester that provides the course		
		DEPT	Department that sets up the course		
		FTIME	Detailed time when student first takes exam		
		RETIME	Detailed time that student retakes exam		
		CREDIT	Course credit		
		SCORE	Student's exam score		
	Scholarship records	SCATEGORY	Scholarship categories	25,589	Educational administration system
		MONEY	The amount of each scholarship awarded		
		SEMESTER	Semester when student obtains scholarship		
		SID	Student identity who obtains scholarship		
	Campus card records	SID	ID of student whose records	52,891,242	Logistics information system
		MEAL	Meal time and consumption amount		
		TTIME	Transportation time in campus		
		DEST	Destination when taking school bus		
Others	Major categories	MID	Unique key for different majors	320	Educational administration system
		DURATION	Duration of achieving a degree		
		DEPT	Department that the major belongs to		
		CATEGORY	Category of different majors		
	Teacher information	TID	Unique key for different teachers	5,337	Teacher
		TNAME	Teacher's name		
		GENDER	Teacher's gender		
		AGE	Teacher's age		
		SSID	Teacher's social security number		
		EDUCATION	Teacher's educational qualification		
		DEGREE	Teacher's degree		
		DEPT	Department that teacher belongs to		
		DEPTID	Department ID that teacher belongs to		

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Xinning Wang received her BS degree and ME degree from Ocean University of China, Qingdao City, China, in 2009 and 2012, respectively, and the PhD degree with the Department of Computer Science and Software Engineering in Auburn University in 2017. She is currently a lecturer with the school of Teaching Center of Fundamental Courses, Ocean University of China, Qingdao, China. Her research interests include spanning data mining and analytics, computer architecture and systems, cloud computing, machine learning and cybersecurity.



Yuben Zhao received the BEng degree in Automation from Shandong Agricultural University, Taian, China, in 2019. She is currently pursuing the MEng degree with the School of engineering, Ocean University of China, Qingdao, China. Her current research interests include machine learning and big data analytics, with applications to educational and oceanic data processing and mining.



Chong Li received his BE degree and ME degree form Ocean University of China, Qingdao City, China, in 2009 and 2012, respectively, and the PhD degree from Auburn University, Auburn, AL, U.S., in 2016. He was a postdoctoral research fellow of the Integrated MEMS Laboratory (IMEMS) of the Center for MEMS and Microsystems Technologies (CMMT) at Georgia Tech from 2016 to 2018. He is currently an associate professor with the Dept. Automation & Measurement, Ocean University of China, Qingdao, China. His research interests include MEMS, control systems, high performance computing, machine learning and artificial intelligence.



Peng Ren received the BEng and MEng degrees in electronic engineering from Harbin Institute of Technology, Harbin, China, and the PhD degree in computer science from the University of York, York, U.K. He is currently a Professor with the College of Oceanography and Space Informatics, China University of Petroleum (East China), Qingdao, China. His research interests include marine information processing, remote sensing, machine learning, etc. Dr. Ren was a recipient of the K. M. Scott Prize from the University of York in 2011 and the Eduardo Caianiello Best Student Paper Award at the 18th International Conference on Image Analysis and Processing in 2015 as a coauthor.